

Torsional behavior of prestressed concrete towers for wind turbines considering the effect of horizontal joint

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ARTICLE INFO

Keywords:

Prestressed concrete tower
Torsional behavior
Grouted horizontal joint
External prestressed steel tendon
Thin-walled concrete structures

ABSTRACT

Torque is a significant partial load born by wind turbine towers. Since the prestressed concrete tower is composed of thin-walled tower segments and horizontal joints, it is prone to torsional failure. However, previous studies on the torsional behavior of prestressed concrete towers are still rare. This paper presents a comprehensive study on the torsional behavior of prestressed concrete towers based on a simplified model, considering the effect of horizontal joint. The cyclic loading test was conducted to provide a preliminary insight of the torsional behavior of the tower. The effects of key parameters on the torque-torsional angle curves were analyzed based on the validated finite element models, summarizing the typical failure modes and corresponding torque-torsional angle curves. Based on the simulated results, the stress states of the tower, the shear force on the horizontal joint, and the prestress variation patterns were revealed, clarifying the torsional failure mechanisms of the tower segment and horizontal joints. Furthermore, the full-process torsional mechanism of the tower was revealed. Lastly, based on the mechanism and variable-angle truss model, calculation formulas for the torsional capacity of the tower segment and the ultimate torque of joint failure were developed, respectively. The practical calculation method of the ultimate torsional capacity was subsequently proposed based on different failure modes, providing a theoretical guidance for engineering calculation and design.

1. Introduction

In recent years, the rapid development of wind power has placed higher demands on towers, which are the primary supporting structures for wind turbines. Traditional steel towers are increasingly facing issues such as vibration and buckling due to their low stiffness. Prestressed concrete towers have become an excellent alternative to steel towers for supporting large wind turbines, because of their advantages of large load-bearing capacity, high stiffness, and lower cost [1]. To facilitate on-site installation, prestressed concrete towers are designed as pre-fabricated structures, as shown in Fig. 1. The thin-walled tower segments are connected to the foundation through grouted horizontal joint without vertical reinforcement passing through. Although the compressive strength of the adopted high-strength grout is high, its bond strength is relatively low, which significantly limits the connection performance of the joint. Thus, prestressing force is required to compress the entire cross-section of the tower, thereby providing a stable boundary condition for the tower. The prestressing force is

generally provided by external post-tensioned steel tendons within the hollow interior of the tower. During the operation of the wind turbine, the load on the prestressed concrete tower mainly comes from the aerodynamic thrust applied to the top of the tower. When the wind turbine performs control operations such as pitch and yaw, the aerodynamic thrust generates both bending moment and torque at the bottom of the tower, making the tower under combined bending-torsion stress state. The magnitudes of the bending moment and torque will change with the variations of the control strategy of the wind turbine, resulting in multiple time-varying load conditions. In torque-dominant load conditions (which occur when the nacelle performs active yawing) [2], the torque is the main load, and the bending moment is relatively small. Combined with the compression applied by gravity and prestress, the tower is mainly subjected to a compression-torsion state, as shown in Fig. 1. In such case, the ultimate state of the tower is governed by its torsional behavior. Furthermore, the torsional behavior forms the foundation for investigating the combined bending-torsion behavior of the tower. In general, the torsional behavior of the

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<https://doi.org/10.1016/j.tws.2025.113610>

Received 6 March 2025; Received in revised form 10 June 2025; Accepted 23 June 2025

Available online 24 June 2025

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prestressed tower should be concerned.

Some studies [3–7] on the prestressed concrete towers have considered the effect of torque in the analysis, but only the elastic stress of the tower based on the torque values under specific loading conditions was verified, and the aim was focusing on optimization design. However, the ultimate torsional behavior of the tower was seldom further investigated. Klein [8] designed a 1:10 scale model of a prestressed concrete tower, considering the effect of torque and conducting non-destructive loading tests. The results indicated that the stress distribution across the horizontal joint under torque was uneven, proving that the current elastic design assumptions do not align with the actual stress conditions. Haar and Marx [9] provided a method for calculating the torsional capacity of horizontal joints. However, they directly used the shear stress calculation formula based on elastic mechanics, without considering the nonlinear effects of material and geometric. Kang [10] conducted numerical simulations on the shear and torsional behavior of tower joints after joint opening and analyzed the impact of joint opening on the shear-torsional capacity. Their study demonstrated the inaccuracy of the calculation method proposed by Haar [9] but did not provide a corresponding calculation method. Chen [11] addressed this issue by proposing a torsional capacity formula. However, the aforementioned studies all assumed torsional failure of horizontal joints during the analysis process, without considering the torsional failure of tower segments. But in fact, when subjected to torque, the ultimate state of the prestressed concrete tower is governed by the coupled effects of joint failure and tower segment torsion. Therefore, the existing studies still has limitations on the torsional behavior of prestressed concrete towers.

Prestressed concrete towers are essentially thin-walled concrete circular columns, so the previous studies on the torsional behavior of concrete members, especially circular members, can provide valuable references for this study. The torsional theories for concrete structures are mainly divided into the skew bending theory [12] and the space truss model [13]. The space truss model proposed by Rausch is the earliest and most classical theory to predict the torsional behavior of concrete members. Torsional theories such as the variable angle truss model [14], the compression field theory [15], and the softened truss model [16,17] have gradually been developed based on the space truss model. In recent decades, subsequent studies on the torsional mechanism of concrete structures based on these theories have been extensively conducted. Jeng [18] presented one new analytical method called Softened Membrane Model for Torsion, considering the strain gradient

of concrete struts in the shear flow zone. Based on this, a unified SMMT theory for hollow and solid concrete members was developed [19]. Stramandinoli [20] proposed a constitutive model that considers the tension-stiffening effect of concrete, which can account for the tensile capacity of concrete between cracks under torsion. Bernardo [21,22] refined the space truss model and proposed one modified variable angle truss model and one generalized softened variable angle truss model, which can predict the entire torque–torsional angle curve of reinforced concrete beams under torsion, including ultimate and cracking states. Additionally, one modified softened truss-model for prestressed concrete beams was presented to consider the effect of prestress on the torsional behavior [23]. Becks [24] developed a novel testing methodology which can apply torsion and compression simultaneously to explore the material behavior before, during, and after macro-cracking, and several thin-walled concrete tubes were tested using this new device. Valipour [25] developed an element for nonlinear torsional analysis of reinforced concrete framed structures, which can consider the effects of normal force, tangential force, and the torsional warping on the torsional stiffness. Ju [26,27] developed the maximum and minimum torsional reinforcement ratio of solid section members based on softened membrane model, correcting the inapplicability of the limit reinforcement ratio in existing design codes. Jeng [28] presented a simplification of the softened membrane model and proposed a unified rational formula for the initial torsional stiffness of solid and hollow concrete members. Kuan [29] presented a sectional analysis tool which can compute the complete nonlinear torsional behavior of reinforced and prestressed concrete members. Jeng [30] conducted torsional tests on prestressed thin-walled hollow concrete beams, comparing the effects of different wall thicknesses and prestress. Besides, the calculation formula for cracking torques of thin-walled hollow prestressed beams was proposed. You [31] proposed an analytical method based on the rotating angle softened truss model, which considers the thickness of the shear flow region in circular concrete members. This method can be used to analyze the structural behavior of members under pure torsion or compression-torsion loads. Mondal [32] proposed an improved softened truss model that incorporates the tension-stiffening effect of concrete. By introducing the tensile contribution of concrete, this model enables more accurate predictions of the torsional behavior for circular members under compression-torsion loads. In recent years, studies of torsional performance have been extended to UHPC members [33–36], FRP-strengthened members [37], and fiber-reinforced members

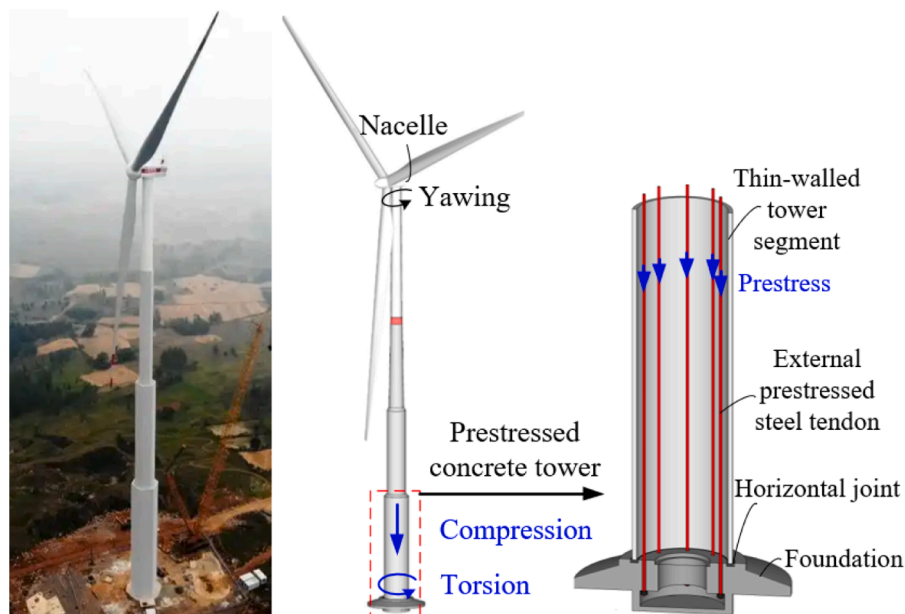


Fig. 1. Schematic of the prestressed concrete tower under compression and torsion.

[38–40]. However, the analysis of torsional capacity is still primarily based on the space truss model. All these studies on the torsional behavior of concrete members can provide important reference on experimental and theoretical investigations of prestressed concrete towers.

In summary, considering the gaps of existing studies on the torsional behavior of prestressed concrete towers and the deficiencies of torsional capacity calculation in current design methods, one comprehensive investigation on the torsional behavior of prestressed concrete towers is presented in this paper based on torsional theories of concrete structures. Firstly, the cyclic destructive loading test was conducted to provide a preliminary understanding of the mechanical behavior of towers under torsion. Subsequently, the finite element (FE) models validated by test results were established to extend the parameter range and comprehensively analyze the full-process torsional mechanism of prestressed concrete towers. Lastly, a practical calculation method for the ultimate torsional capacity of the tower was proposed based on the torsional mechanism. This study can provide theoretical guidance for calculation and design of wind turbine prestressed concrete towers under torque-dominant load conditions.

2. Torsional test program

2.1. Test design and process

Appropriate simplifications were adopted for the research objective to more clearly obtain the torsional behavior of the tower considering horizontal joints. As the reinforcement ratio and material strength of the traditional vertical joints are higher than those of the tower segments, the vertical joints can be considered as cast-in connection [4], thus the influence of the vertical joints was ignored to simplify the specimen fabrication and to focus on the torsional performance under horizontal joint effects. When the heights of each tower segment in the tower tube are the same, the bottom joint will play a controlling role [41,42]. Therefore, only the most critical joint between the tower bottom and the foundation is considered. According to the simplification rules, the specimen dimensions were scaled based on the bottom concrete tower segment of the hybrid tower, with a diameter of 610 mm and a wall thickness of 55 mm. Based on the previous study [43], the specimen height was 2600 mm, including a 200 mm transition section designed to anchor the prestress tendons and connect the loading device, as illustrated in Fig. 2(a). The reinforcement configuration of the specimen was

based on the actual engineering design, using double-layer and bidirectional reinforcement to form a framework. The reinforcement diameter was 4 mm. The vertical and horizontal reinforcement ratios were 1.00 % and 0.61 %, respectively. The reinforcement framework is shown in Fig. 2(b). To simulate the grouted horizontal joint between the tower base and the foundation, a solid concrete base with a height of 300 mm and a diameter of 1200 mm was designed. A groove was reserved at the top of the base, which was filled with high-strength grouting material. After grouting, the specimen was placed on the top of the groove. Then the horizontal joint was completed once the grout was formed, as shown in Fig. 3. The specimen and the base were only connected by the adhesion of the grouting interface, but the connection can be reliable after applying prestress. The design prestress was scaled based on the axial compression ratio induced by prestressing in the engineering, which is represented as prestress ratio n in this study. A total of eight externally unbonded prestressed tendons were evenly distributed along the cross-section. The tendons were anchored at the top of the transition section and the bottom surface of the concrete base, as shown in Fig. 2 (a). The designed prestress ratio n is 0.22. After tensioning, the measured applied prestressing force is 863.68 kN, with an average tensile stress of 771.15 MPa, resulting in an actual prestress ratio of 0.228.

The compressive strength and elastic modulus of the concrete and grouting material were tested according to the specifications of GB/T50081–2002 [44], with the results shown in Table 1. Additionally, the yield strength, ultimate strength, and elastic modulus of the steel reinforcement bar and steel strand used for prestressed tendons were tested in accordance with GB/T228.1-2010 [45], with the results shown in Table 2.

2.2. Test setup and procedures

A comprehensive loading device was used to apply load to the specimen. The loading device primarily consists of a loading beam, a cross hinge, an actuator, an anchoring steel cover, and a steel base, as shown in Fig. 4. The steel base was fixed to the ground, while the concrete base was connected to the steel base using high-strength bolts. Bolts were embedded at the top of the specimen, to allow the specimen connected to the steel cover. Additionally, the prestressed tendons were anchored to the steel cover to apply the prestressing force to the specimen. The top surface of the steel cover was connected to the bottom of the loading beam. One end of the actuator was connected to the reaction wall, while the other end was connected to the loading beam through a

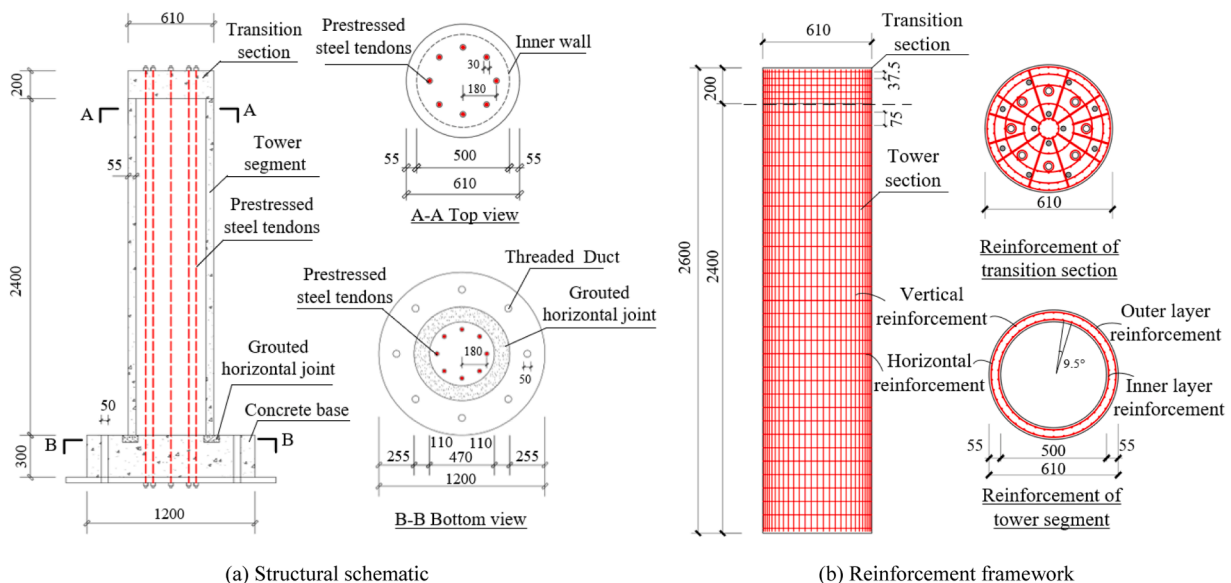


Fig. 2. Test specimen.

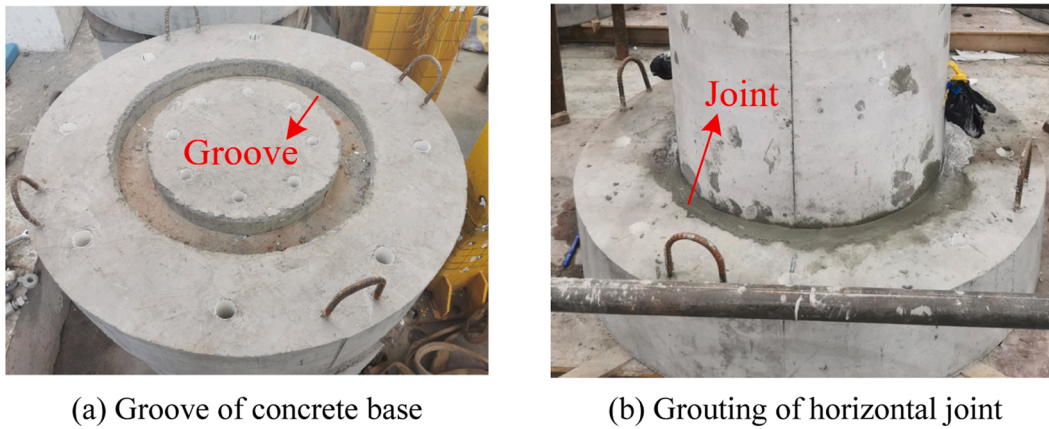


Fig. 3. Construction process of the horizontal joint.

Table 1

Material properties of concrete and grout.

Material type	Concrete for specimen	Concrete for base	High-strength grout
Compressive strength	47.27 MPa	69.81 MPa	83.52 MPa
Elastic modulus	32,687.67 MPa	37,528.25 MPa	39,767.30 MPa

Table 2

Material properties of steel.

Material type	Yield strength f_y (Mpa)	Ultimate strength f_u (Mpa)	Elastic modulus E_s (Mpa)
Steel reinforcement bars	523.18	580.65	199,325.2
Steel strand	1804.86	1996.43	204,021.3

cross hinge. The cross hinge was assembled from two directional ear plates connected by a cross pin, with a rolling sleeve and a limiting device installed between the pin and the ear plates, ensuring that the rotation at the top of the specimen was not restricted. The actuator was installed at the left end of the loading beam, and a rigid rod was installed at the center of the loading beam, as shown in Fig. 4. The rigid rod has a high stiffness and can be considered to experience no deformation. Once the rod was connected, the horizontal displacement of the specimen was restricted, and the force from the actuator can only apply a torque T to the specimen, without inducing a bending moment. Combined with the prestressing force P applied by the steel tendons, the compression-torsion loading was achieved. The loading mode is shown in Fig. 4(b).

The geometric relationship of the device under torsion is shown in Fig. 4(c). θ is the torsional angle of the specimen, Δ is the horizontal displacement of loading point, B is the distance from the loading point and the center of the loading beam (i.e., the length of the torsion arm), L_0 is initial length of the actuator, C is distance from the center of the loading beam to the end of the actuator, F is the horizontal force, and β is the angle between the actuator and the loading beam. According to previous studies [46,47], the torsional angle θ is generally linearly related to the horizontal displacement Δ . The torsional angle θ and the torque T can be calculated using following equations:

$$\theta = \arcsin(\Delta/B) \quad (1)$$

$$T = FB\sin\beta \quad (2)$$

$$\text{With, } \sin\beta = \frac{C}{\Delta + L_0} \sin(\theta + \alpha)$$

$$\alpha_0 = \arctan(L_0/B)$$

During loading, the horizontal force F was applied by the actuator can be recorded by a load sensor, and the horizontal displacement Δ at the loading point can be measured using a high-precision linear displacement transducer (LVDT). The torque T and torsional angle θ can be obtained by converting the measured values of F and Δ according to Eqs. (2) and (1), respectively. The measurement arrangement is shown in Fig. 5. The prestressed steel tendons were tensioned according to the designed prestress value, followed by the application of the torque loading. The displacement-controlled loading was adopted, and the displacement increment for each loading step was set at +2 mm and -2 mm for positive (+) and negative (-) directions, respectively. This displacement increment corresponded to a torsional angle increment of $\pm 0.143^\circ$. Each displacement was applied once in each loading step. After reaching the corresponding displacement of each step, the load was maintained for 2 min to facilitate data collection, observe the phenomena, and allow the specimen's deformation to stabilize. The test was stopped when the load dropped below 85 % of the peak load or when significant failure characteristics of the specimen were observed.

2.2. Test phenomenon and failure modes

The failure mode of the specimen is shown in Fig. 6. It can be observed that the damage occurred in the middle section of the specimen with significant torsional deformation. In the early loading stage, a long clockwise diagonal crack developed on the upper part of the specimen, extending from the transition section to the middle height of the specimen. As loading progresses, more short diagonal cracks appeared near the initial crack. These cracks were generated by both positive and negative loading directions, and intersected with each other. During positive or negative loading, the reverse diagonal cracks could totally close. After the specimen yielded, the torsional angle increased significantly, and the load could continue to rise slowly. At this state, the initial diagonal crack had developed into a through-thickness helical crack, extending from the top to the bottom of the specimen. As the torsional angle continued to increase, the diagonal cracks further expanded, eventually covering the entire specimen height. This resulted in the formation of dense X-shaped intersecting helical cracks, as shown in Fig. 6. Subsequently, the concrete cover in the middle section of the specimen fell off, and the cracks in the middle became significantly wider. The internal reinforcement was visibly subjected to torsional deformation, and the concrete on sides of the diagonal cracks showed significant compressive bulged. At this state, the specimen had experienced significant torsional deformation, but the torsional angle could continue to increase while the load remained nearly constant. At the next level of positive torsional angle after

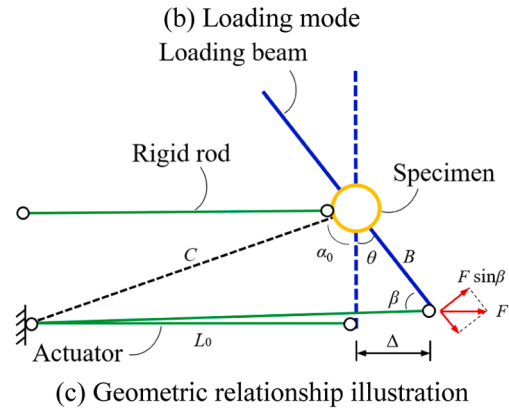
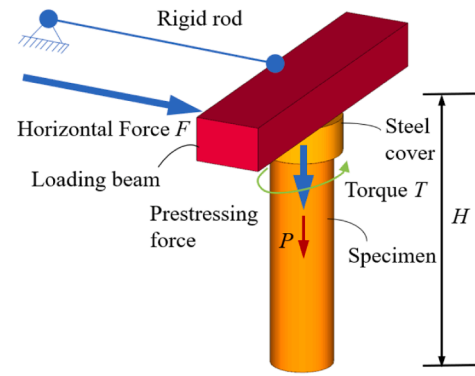
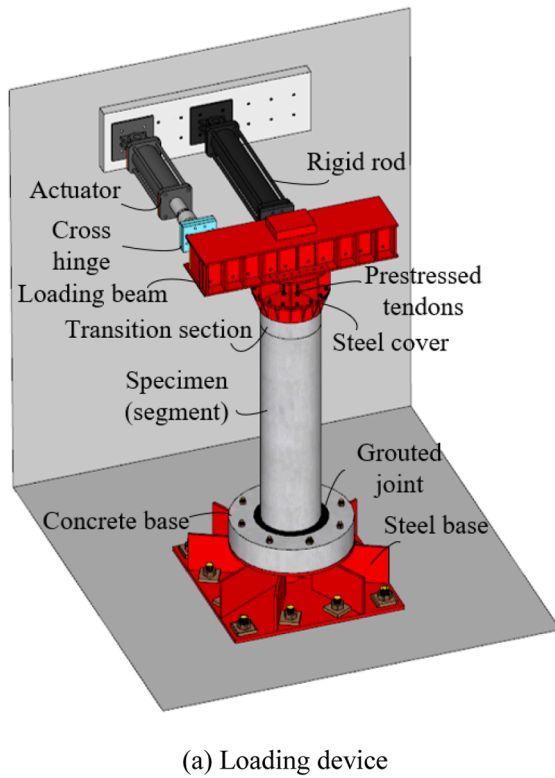


Fig. 4. Test setup.

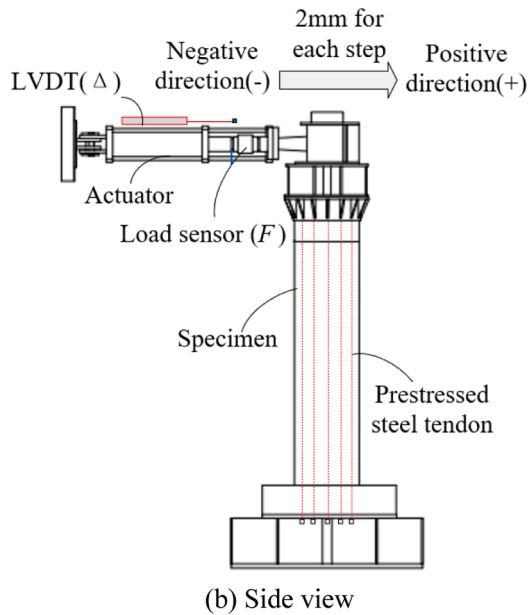
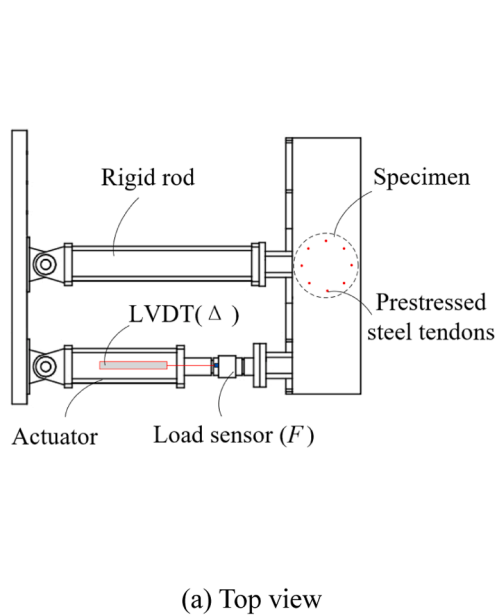


Fig. 5. Measurement arrangement.

reaching the peak load, concrete crushed at the west side of the middle section and the twist of the reinforcement could be clearly observed within the crack gaps. The positive load decreased to below 85 % of the peak load but could still be sustained. During the Subsequent negative loading stage, the diagonal cracks in the middle section rapidly expand. A large area of concrete crushed, with severe twist of the reinforcement. The upper and lower parts of the specimen became misaligned due to excessive torsional deformation. The load decreased rapidly and could not be stabilized, leading to the final failure. After failure, no significant

torsional sliding was observed at the horizontal joint. Therefore, the failure mode can be summarized as: the specimen developed dense diagonal cracks, with severe twist of the reinforcement within the cracks, and concrete on sides of the main diagonal crack crushed.

2.3. Load–displacement curves

The hysteresis curve and skeleton curve of the specimen is shown in Fig. 7. Except for the first loading circle, the hysteresis loops of the

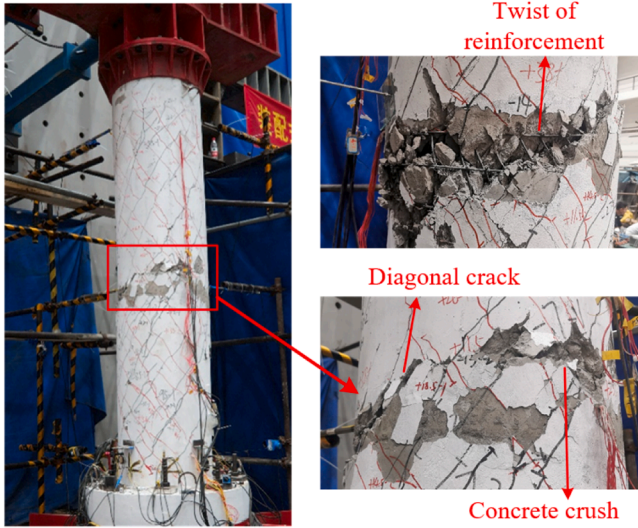


Fig. 6. Test failure mode.

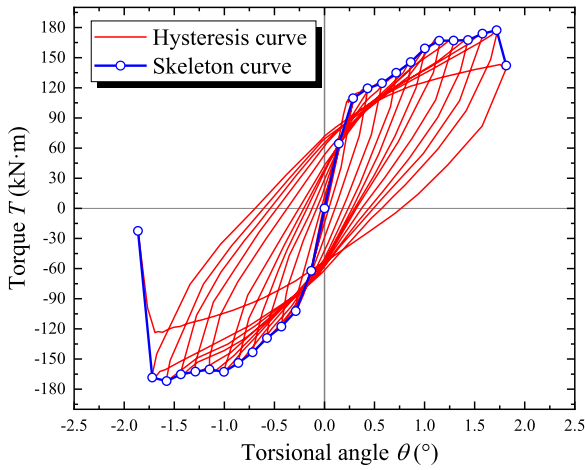


Fig. 7. Hysteresis curve and skeleton curve.

subsequent loading stages all showed a significant enclosed area, indicating that the specimen exhibited irrecoverable deformation in the early stages. As the torque increased, the crack of concrete cracks intensified. The stiffness of the hysteresis loops significantly decreased, and the enclosed area expanded notably, with a marked increase in residual displacement. This indicated that the reinforcement yielded and gradually developed plasticity. In the later loading stages, the skeleton curve remained horizontal line, and the load remained almost constant, showing a significant plastic stage. Combined with the experimental observations, this indicated that the concrete had undergone severe cracking, with noticeable torsional deformation of the reinforcement. This suggested that the contribution of the reinforcement allowed the tower to undergo extended plastic deformation, thus maintaining its load-bearing capacity. At the ultimate displacement, the enclosed area of the hysteresis loop increased significantly, and the load dropped sharply, leading to the failure of the specimen.

2.5. Stiffness degradation

The secant stiffness K_i was introduced to analyze the stiffness degradation pattern, and the secant stiffness are showed in Fig. 8. The secant stiffness K_i is determined by:

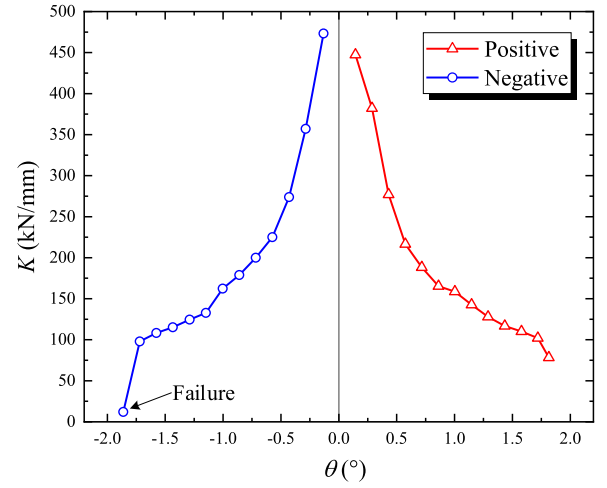


Fig. 8. Stiffness degradation curve.

$$K_i = \frac{|+T_i| + |-T_i|}{|+\theta_i| + |-\theta_i|} \quad (3)$$

Where, $+T_i$ and $-T_i$ are the peak torques in the i th loading stage of positive and negative direction, respectively; $+\theta_i$ and $-\theta_i$ are the torsional angles corresponding to $+T_i$ and $-T_i$.

It can be found that the positive and negative stiffness degradation curves were nearly symmetric. The stiffness degraded significantly before the torsional angle reached $\pm 0.5^\circ$. As the torsional loading progressed, the concrete gradually cracked and the tensile stress shifted from the concrete to the reinforcement, resulting in a sharp degradation of the torsional stiffness. After this point, the crack development gradually stabilized, and the reinforcement began to yield, thus the specimen undergone plastic deformation at an early stage. As a result, the stiffness degradation curve became more gradual. The negative stiffness degradation curve showed a sharp drop at the ultimate displacement, because at this point, the concrete reached the ultimate compressive strain under compression-torsion load and suddenly crushed, leading to a rapid decrease in the specimen's load.

2.6. Energy dissipation and residual displacement

The enclosed loop area in the hysteresis curve can represent the energy dissipation capacity of specimens. The equivalent viscous damping coefficient h_e [48] is used to quantify the energy dissipation capacity:

$$h_e = \frac{1}{2\pi} \frac{S_1}{S_2} \quad (4)$$

Where, S_1 is the enclosed area of the hysteresis loop, representing the total energy dissipated during that loading cycle; S_2 is the nominal elastic potential energy corresponding to the maximum displacement point of that loading cycle.

From Fig. 9, it can be observed that the equivalent viscous damping coefficient of the specimen increased rapidly in the early stages of loading. After the concrete cracked under torsion, the reinforcement began to carry tensile stress. The process of concrete cracking and reinforcement yielding quickly increased the energy dissipation. After the torsional angle reached $\pm 0.5^\circ$, the equivalent viscous damping coefficient increased at a slower rate, indicating that the reinforcement has progressively yielded. From this point onward, the energy dissipation at each loading stage increased uniformly. The positive equivalent viscous damping coefficient remained around 15 after this point, until a significant increase occurred when local concrete spalled at the ultimate displacement. The negative equivalent viscous damping coefficient

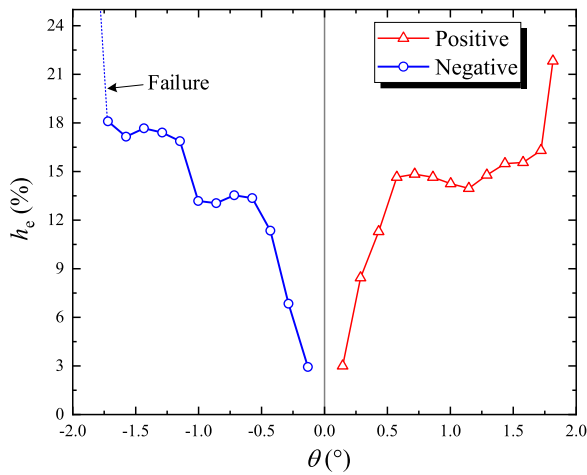


Fig. 9. Equivalent viscous damping coefficient curve.

increased significantly after the torsional angle reached -1° . Based on the experimental observations, large cracks appeared in the middle section at this loading stage, and the vertical reinforcement experienced severe torsional deformation, leading to substantial plastic deformation and energy dissipation. Afterward, the equivalent viscous damping coefficient stabilized again until the ultimate displacement, when the complete crushing of the concrete caused a sharp increase in energy dissipation.

The residual displacement is the irreversible displacement of the specimen when unloaded to zero at each loading cycle. The positive and negative residual torsional angles θ_r are shown in Fig. 10. The residual displacement increased rapidly with the torsional angle in the early loading stages, indicating that due to the torsional cracking of the concrete, the specimen quickly undergone irreversible torsional deformation. After this point, the residual displacement increased at a nearly constant rate. When the torsional angle reached $\pm 1^\circ$, the rate of increase in residual displacement became larger. Combined with the skeleton curve, at this state, the peak torque has been reached, the reinforcement has largely yielded, and plastic deformation has developed significantly, resulting in more irreversible deformation. At the ultimate displacement, due to concrete crushing, the specimen experienced significant damage, and the residual displacement increased sharply again.

Combining the analysis of energy dissipation and residual displacement, it can be concluded that when the prestress ensures a stable connection of the joint, the prestressed concrete tower exhibits significant plastic deformation under torsion. The self-resetting capability of

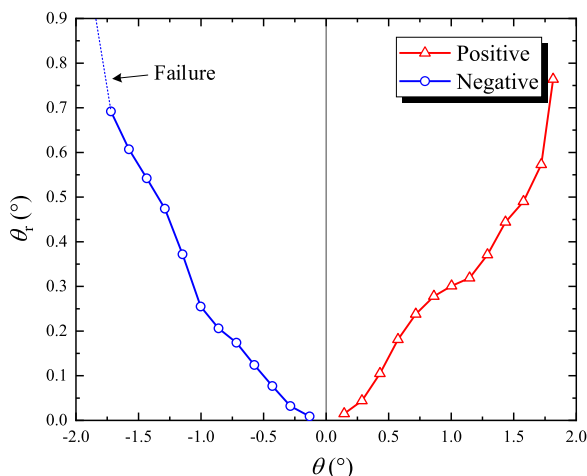


Fig. 10. Residual displacement.

the external prestressed structures [49–51] cannot be utilized, and the post-damage reparability of the structure is low. However, due to the plasticity from reinforcement yielding, the tower still retains some degree of ductility.

3. FE modelling and parametric analysis

3.1. FE model description and validation

Based on the test results, the torsional mechanism of the prestressed concrete tower was further investigated through finite element (FE) analysis. The general FE analysis software ABAQUS was adopted.

The prestressed concrete tower model consists of the tower segment, reinforcement framework, concrete base, and steel tendons, as shown in Fig. 11. The tower segment and concrete base were modeled using C3D8R solid elements, while the reinforcement framework and steel tendons were modeled by T3D2 truss elements. Through mesh sensitivity analysis, the solid element size for the tower segment and base was set to 40 mm, which could ensure accuracy while improving computational efficiency. The reinforcement framework was embedded within the tower segment. Tie constraints were applied between the top plate and the transition section, as well as between the bottom plate and the concrete base. The two ends of the prestressed steel tendons were connected through coupling constraints to the anchorage regions at the top plate and the bottom plate, respectively. Since the reinforcements at the bottom of the tower segment do not pass through the horizontal joint and the bond performance of the joint is weak, the connection between the tower segment effect is mainly ensured by the prestress. Therefore, the surface-to-surface contact was used to simulate the interaction behavior between the bottom of the tower segment and the horizontal joint. In the normal behavior, hard contact was adopted, while the tangential behavior was simulated using the Coulomb friction model defined by the penalty function method. The friction coefficient μ was set to 0.9. The combination of these two methods can effectively simulate the compression transferred to the joint by the prestress, as well as the interface frictional resistance generated as a result. Besides, the bottom surface of the bottom plate was totally fixed in the model. A reference point was set at the loading height, and coupled with the top plate. Then, the torsional angle was applied at the loading point, as shown in Fig. 11. After coupling both ends of the steel tendons, a temperature field was applied to the tendons in the predefined fields to apply the prestress.

The plastic damage model in ABAQUS was adopted to simulate the material behavior of concrete. This model can effectively describe the unloading stiffness degradation and reloading stiffness recovery characteristics of concrete structures under cyclic loading. The stiffness degradation was controlled by the damage factor d , and d was calculated using the damage model presented by Wang and Yu [52]. The uniaxial compressive stress-strain curve of concrete was described by the classical Hognestad curve [53], which consists of a parabolic ascending stage and a linear descending stage. The tensile softening behavior of concrete was simulated using the fracture energy method in the CDP model. The tensile peak stress and fracture energy are calculated by the methods in [54] and [55], respectively. For the steel reinforcement, the stress-strain relationship was described using the three-stages model suggested in GB50010–2010 [56]. Since the prestressed tendons were primarily in the elastic stage during the loading process and their post-yield hardening behavior could be neglected. Therefore, the isotropic ideal elastoplastic model was adopted to describe their stress-strain behavior. The parameters of strength and elastic modulus of each material were input based on the results of Tables 1 and 2.

The FE model was validated by experimental results. The comparison between the measured and simulated failure mode of the specimen is shown in Fig. 12. It can be observed that in the FE model, significant torsional deformation occurred in the tower segment, with a noticeable misalignment between the upper and lower sections at the torsional

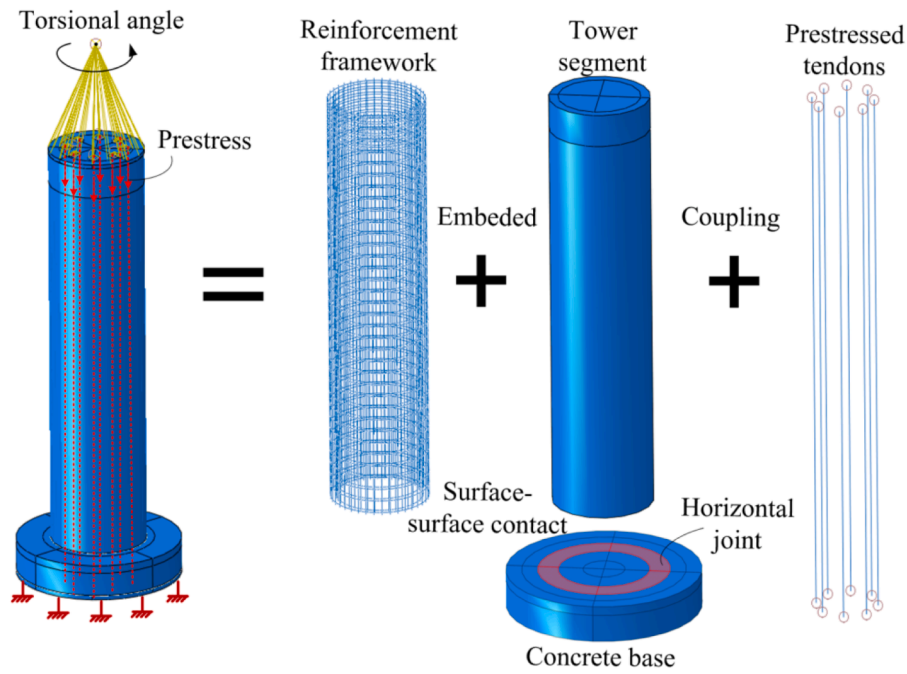


Fig. 11. FE model of prestressed concrete tower.

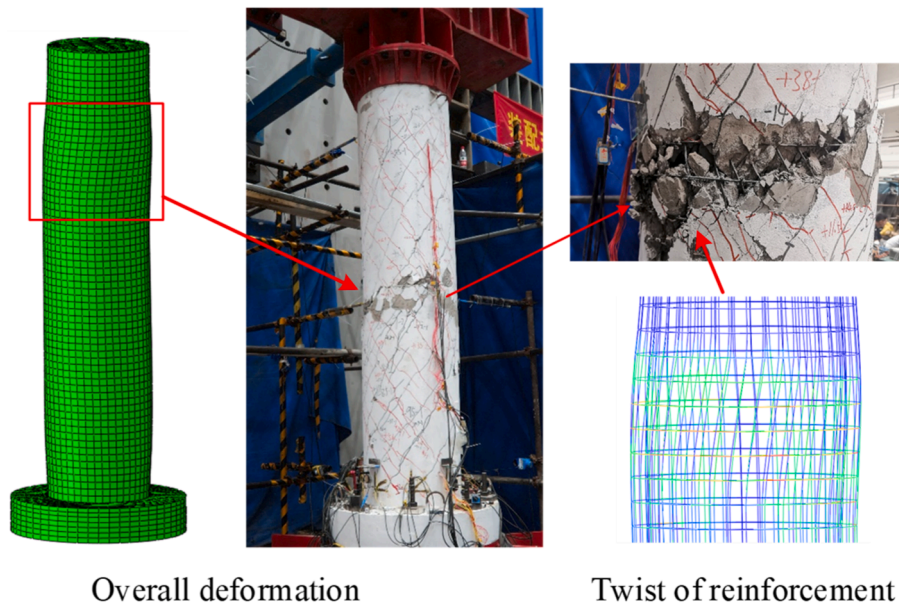


Fig. 12. The comparison between the measured and simulated failure mode.

location. At the same time, the reinforcement in this area experienced severe twist, and the plastic deformation occurred on a large area, which is consistent with the experimental observations. The most severe torsional location predicted by the model differed from the test. This can be attributed to that the torque applied along the tower segment was the same except for the boundary area, making it difficult to precisely predict the region where torsional deformation was most severe. Overall, the FE model could still simulate the typical failure mode well, where the tower segment and reinforcement undergone significant torsion, which is consistent with the test results.

The measure and simulated $T-\theta$ curves were compared in Fig. 13. Although there were some differences in the shape of the hysteresis loops, the initial stiffness and ultimate capacity of the measured and simulated hysteresis curves were in good agreement. From Fig. 13(b), it

can be further observed that the trends of the measured and simulated skeleton curves were consistent. Both exhibited a complete linear-elastic and elastoplastic phase, with the ultimate capacities being well agreement. The simulated positive and negative average ultimate capacity was 169.94 kN·m, with an error of 2.76 % compared to the measured average value, indicating a low error. As can be found from the comparisons of the failure modes, skeleton curves, and ultimate capacity, the FE model can well predict the bearing performance of the tower. Given that this study primarily focuses on the torsional bearing performance of prestressed concrete towers, the established model demonstrates high accuracy and can be used for further analysis in subsequent analysis.

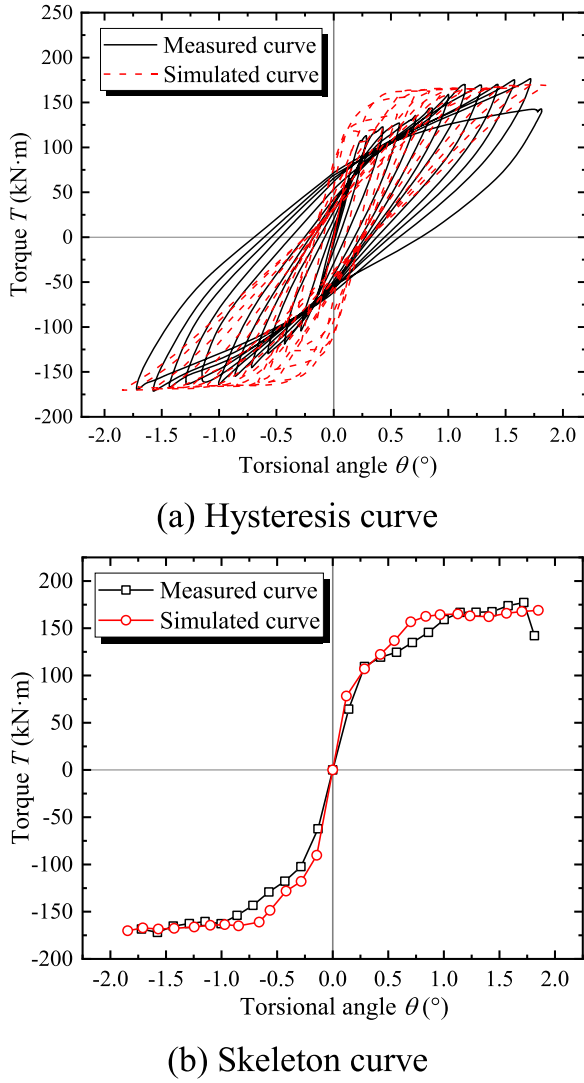


Fig. 13. The comparison between the measured and simulated T - θ curves.

3.2. Parametric analysis

3.2.1. Overview

Considering the limited experimental parameter range, a parametric analysis was conducted based on the validated FE model to further investigate the effects of key parameters such as prestress ratio, hollow ratio on the torsional behavior of prestressed concrete towers. To make the parameter variations more representative, full-scale models were used for the analysis. The dimensions of the basic model were based on the bottom section of a H146–3.4MW-140 m prestressed concrete tower. The tower height H is 30,000 mm, the outer diameter D is 7000 mm, and the thickness t is 400 mm, with a hollow ratio of 0.89. Sixteen bundles of external unbonded prestressed tendons are uniformly arranged along the section, with each bundle consisting of 34 steel strands of 15.2 mm in diameter and a strength grade of 1860 MPa. The total initial prestressing force is 92.92 MN, and the design prestress ratio n is 0.2. The double-layer and bidirectional reinforcement arrangement is adopted, and the strength grade is HRB400 [56]. The vertical reinforcement ratio is 1.31 %, and the horizontal reinforcement ratio is 0.99 %. In cases where vertical loads were not applied, the prestress ratio n was equivalent to the axial compression ratio m . During the parametric analysis, all parameters remained unchanged as specified above except for the varying ones.

3.2.2. Effect of hollow ratio χ

Hollow ratio is the proportion of the hollow area relative to the total cross-sectional area, which is a critical parameter of the tower section. The outer diameter was kept constant, and the thickness of the tower was varied to change the hollow ratio, with the hollow ratios ranging from 0.8 to 0.95. The torque - torsional angle curves (T - θ curves) for tower with different hollow ratios are shown in Fig. 14(a). Due to the increase in section modulus of torsion, both the initial stiffness and the ultimate capacity are significantly improved for the towers with small hollow ratio. The ultimate capacity of the tower with a hollow ratio of 0.8 is 50.21 % higher than that of the tower with a hollow ratio of 0.95. Furthermore, the T - θ curves of the towers with small hollow ratios show a more gradual increase in torque after the stiffness change. This is due to the reduction in axial compression ratio of the tower with a smaller hollow ratio, given that the total prestress remains unchanged. In contrast, the towers with large hollow ratios show faster load increase after the stiffness change, and once the torsional capacity is reached, the torque rapidly decreases. In addition, when the hollow ratio is less than 0.83, the T - θ curve experiences only a small increase after the stiffness change before becoming nearly horizontal. This may be attributed to that the decrease in axial compression ratio leads to that the prestress cannot provide enough friction resistance at the horizontal joint to bear the torque. As a result, the joint begins to slip before the torsional failure of the tower segment, causing the boundary conditions of the tower to fail and transforming into a movable mechanism that rotates along the joint interface. Consequently, the torsional angle can continue to increase while the torque remains almost unchanged.

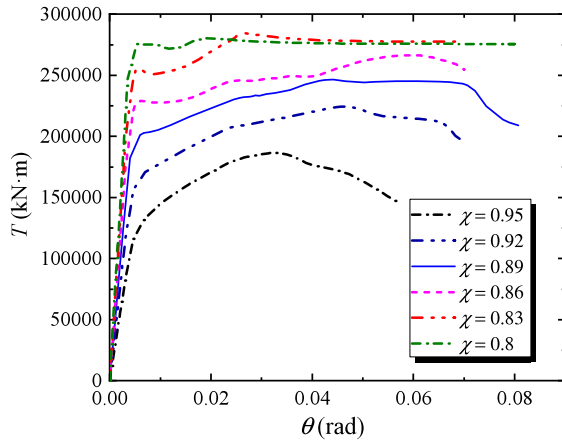
It is worth noting that the tower with a hollow ratio of 0.83 has a greater torsional capacity than the tower with a hollow ratio of 0.8. This is because, when the total prestress remains constant, the tower with a larger thickness has a smaller torsional shear force in unit area provided by friction at the joint interface. Therefore, the ultimate torque of joint failure is smaller. However, if the prestress ratio is kept constant while varying the hollow ratio (as shown in Fig. 14(b)), the torsional capacity significantly increases as the hollow ratio decreases. The ultimate capacity of the tower with a hollow ratio of 0.8 is 119.49 % higher than that of the tower with a hollow ratio of 0.94, which also highlights the significant impact of the prestress ratio on the torsional capacity.

3.2.3. Effect of diameter D

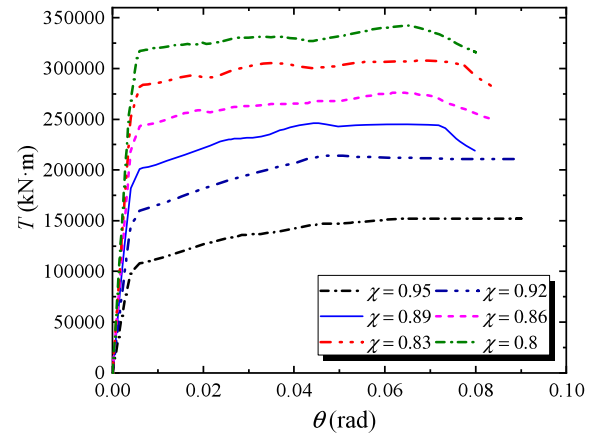
The thickness and other parameters were kept constant, and the diameter of the tower was varied within the range of 4–10 m, as shown in Fig. 15. The tower height is proportionally scaled to maintain the constant height-diameter ratio when changing the diameter. As can be seen, the torsional capacity of the tower with a diameter of 10 m is enhanced by 426 % compared to the tower with a diameter of 4 m. The increase in diameter significantly enhances the section modulus of torsion, leading to a substantial improvement in both the initial stiffness and ultimate capacity. Additionally, the increase in diameter leads to a decrease in the axial compressive ratio. The reduction in axial compression ratio improves the torsional ductility of the larger diameter tower. Both the 8.5 m and 10 m diameter towers exhibit a failure mode due to joint failure, with the tower segment not yet reaching the torsional capacity. In such cases, further increasing the prestress can fully utilize the tower's torsional capacity.

3.2.4. Effect of height-diameter ratio H/D

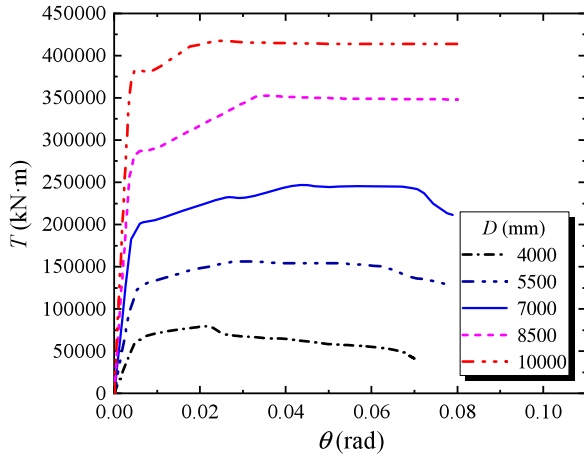
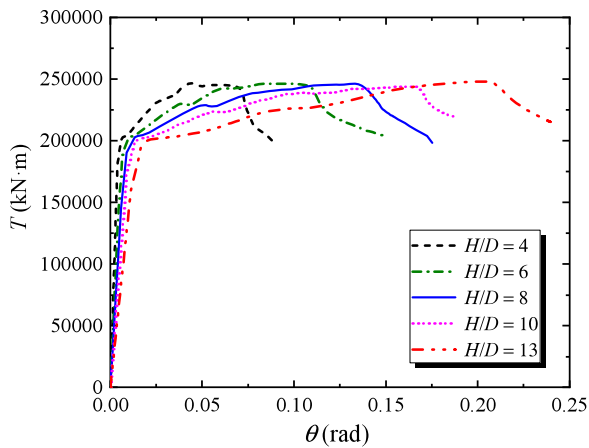
The height-diameter ratio of the tower is varied within the range of 4–13, corresponding to tower heights ranging from 28 m to 91 m, as shown in Fig. 16. It can be observed that the torsional capacities of towers with different height-diameter ratios are very close, with a maximum difference of only 1.69 %, indicating that the height-diameter ratio has no significant impact on the torsional capacity. This may be attributed to that the ultimate capacity is solely determined by the section's torsional behavior for unrestrained torsion. As long as the height of the tower is sufficient to move the critical torsional region



(a) Keeping prestress constant



(b) Keeping prestress ratio constant

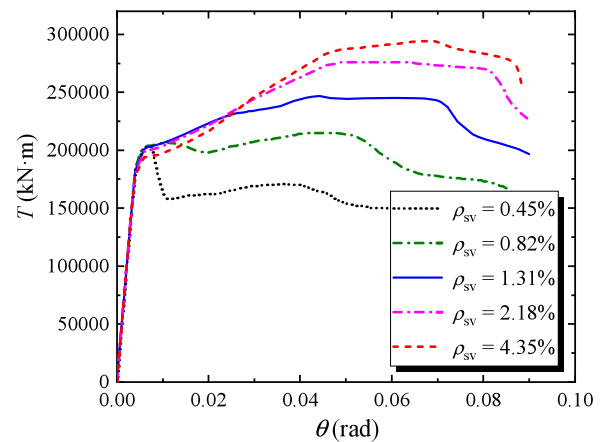
Fig. 14. T - θ curves of prestressed concrete towers with different hollow ratios.Fig. 15. T - θ curves of prestressed concrete towers with different diameters.Fig. 16. T - θ curves of prestressed concrete towers with different height-diameter ratios.

away from the constrained area near the joint, allowing the tower to undergo unrestrained torsion, the height will no longer significantly affect the torsional capacity of the tower. This further proves the universality of the height-diameter ratio selected for the specimen and the basic model in this study. However, the stiffness of the large height-

diameter ratios towers is significantly smaller, and their torsional ductility is notably enhanced. This is attributed to that the torsional deformation of the tower is governed by torsional shear strain, which is determined by the sectional dimensions and the unit-length torsional angle. In the parametric analysis, a uniform torsional angle was applied at the tower top. Thus, compared to towers with smaller height-diameter ratios, those with larger height-diameter ratios require a greater overall torsional angle to achieve the same unit-length torsional angle, thereby generating the same magnitude of shear strain. Therefore, towers with larger height-diameter ratios experience significantly larger torsional angles but show close torsional capacities.

3.2.5. Effect of reinforcement ratio

The spacing of vertical reinforcement were varied at 750 mm, 500 mm, 250 mm, 150 mm, and 75 mm while keeping the diameter constant, and the corresponding reinforcement ratios ρ_{sv} were 0.45 %, 0.82 %, 1.31 %, 2.18 %, and 4.35 %. The T - θ curves of towers with different vertical reinforcement ratios are shown in Fig. 17. It can be observed that increasing the vertical reinforcement ratio effectively enhances the torsional capacity of the tower. The tower with a vertical reinforcement ratio of 2.18 % demonstrates a 37.37 % improvement in torsional capacity compared to the tower with a reinforcement ratio of 0.45 %. For the tower with a vertical reinforcement ratio of 0.45 %, its T - θ curve shows only a brief load increase after reaching the cracking torque before quickly reaching the ultimate torque, followed by a sharp decline in capacity. This indicates poor ductility and exhibits a typical under-

Fig. 17. T - θ curves of prestressed concrete towers with different vertical reinforcement ratios.

reinforced failure. Therefore, increasing the vertical reinforcement ratio can improve the torsional ductility but has minimal effect on the cracking torque.

The spacing of horizontal reinforcement were varied at 750 mm, 500 mm, 250 mm, 150 mm, and 75 mm while keeping the diameter constant, and the corresponding reinforcement ratios ρ_{sh} were 0.34 %, 0.62 %, 0.99 %, 1.65 %, and 3.28 %. The T - θ curves of towers with different horizontal reinforcement ratios are shown in Fig. 18. The torsional capacity of the tower with a horizontal reinforcement ratio of 1.65 % is 45.12 % higher compared to that with a reinforcement ratio of 0.34 %. Compared to increasing the vertical reinforcement, the improvement of horizontal reinforcement on torsional performance of tower is more pronounced. It can also be observed that the torsional capacity of the tower with a reinforcement ratio of 3.28 % is similar to that of the tower with a reinforcement ratio of 1.65 %. This may be attributed to that when the reinforcement ratio is excessively high, the torsional performance of the tower segment is significantly enhanced. In this case, the torsional capacity is governed by the ultimate torque of the joint interface. On the other hand, increasing the horizontal reinforcement significantly improves ductility but has no impact on the cracking torque. This indicates that the cracking torque is primarily determined by the cross-sectional dimensions of the tower segment and the concrete strength, with minimal influence from the reinforcement configuration.

3.2.6. Effect of reinforcement strength

The effect of reinforcement strength was analyzed by varying the reinforcement strength from 235 MPa to 500 MPa. The T - θ curves of towers with different reinforcement strength are shown in Figs. 19 and 20. The torsional capacity of the tower with a horizontal reinforcement strength of 500 MPa increased by 14.97 % compared to that with a horizontal reinforcement strength of 300 MPa. In contrast, the torsional capacity of the tower with a vertical reinforcement strength of 500 MPa increased by only 5.57 % compared to that with a vertical reinforcement strength of 300 MPa. The strength of the vertical reinforcement was not fully utilized, whereas enhancing the horizontal reinforcement strength is more beneficial for improving the torsional capacity of the tower, which is consistent with the conclusion of the reinforcement ratio series.

3.2.7. Effect of configuration of steel tendons

The number of steel strands for each tendon was varied from 30 to 110, while keeping the bundle number of tendons at 16 and the total prestress ratio at 0.2. The T - θ curves for towers with different numbers of steel strands are shown in Fig. 21. In addition, the bundle number of tendons was distributed evenly across 4 to 40 within the cross-section, while maintaining a prestress ratio of 0.2. The T - θ curves of towers with different bundle numbers of steel tendons are shown in Fig. 22. It

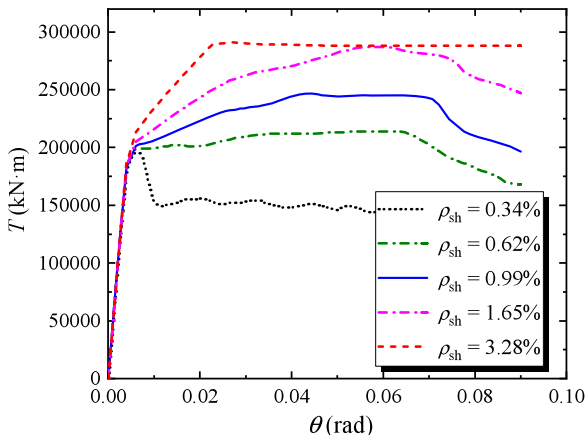


Fig. 18. T - θ curves of prestressed concrete towers with different horizontal reinforcement ratios.

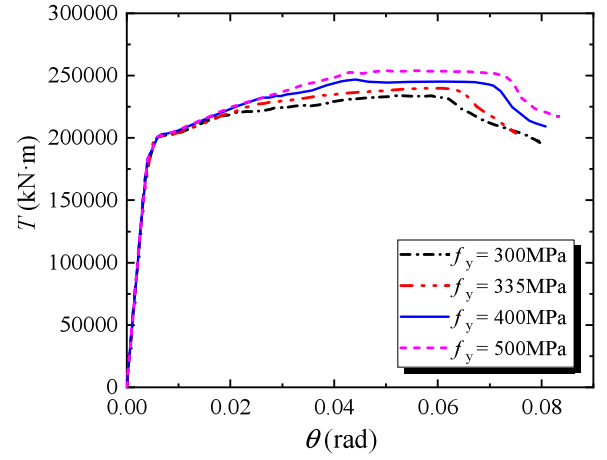


Fig. 19. T - θ curves of prestressed concrete towers with different vertical reinforcement strength.

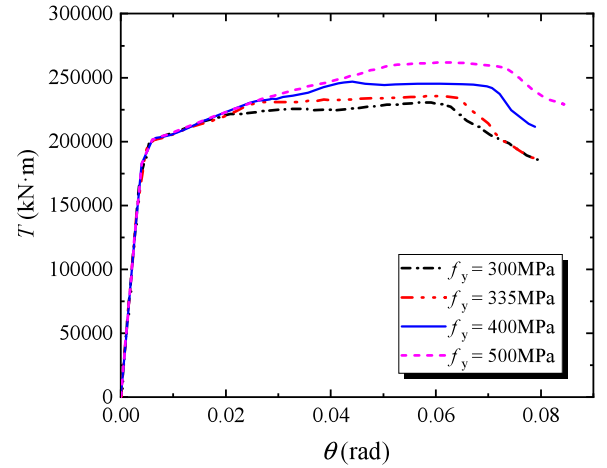


Fig. 20. T - θ curves of prestressed concrete towers with different horizontal reinforcement strength.

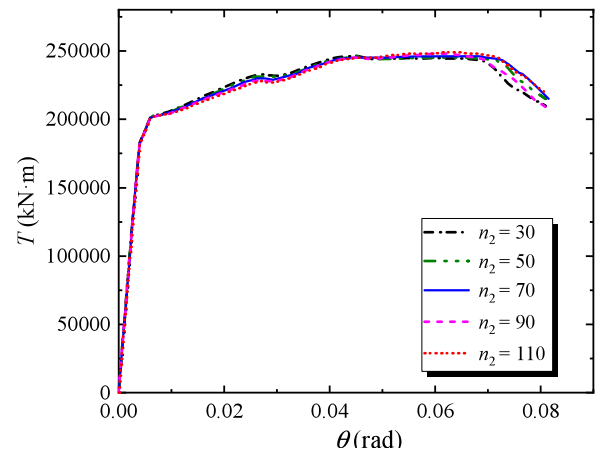


Fig. 21. T - θ curves of prestressed concrete towers with different numbers of steel strands.

can be observed that, with the total prestress remaining constant, neither changing the number of distributed bundles nor altering the number of strands can significantly impact the torsional performance of the tower. The T - θ curves are almost identical. This indicates that the

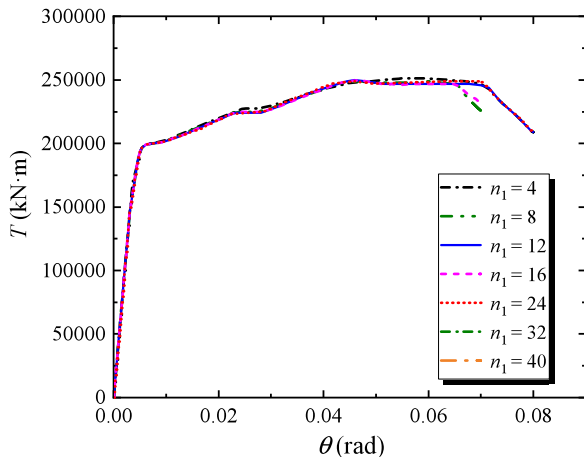


Fig. 22. T - θ curves of prestressed concrete towers with different bundle numbers of steel tendons.

steel tendons do not directly contribute to torsional resistance or generate significant torsional deformation. Their contribution to the torsional capacity of the tower primarily originates from the prestressing force. The prestressing force in towers with different prestress configurations can be considered equivalent.

3.2.8. Effect of prestress ratio

From the results of the aforementioned analysis results, it can be observed that the axial compression ratio of the tower is a key parameter influencing the torsional behavior, with most of it contributed by the prestress applied by the steel tendons. Therefore, the prestress ratio was increased from 0.1 to 0.7, and the T - θ curves of towers with different prestress ratios are shown in Fig. 23. It can be observed that the T - θ curves of towers with different prestress ratios show significant differences. When the prestress ratio is below 0.6, the torsional capacity of the tower increases significantly with the growth of the prestress ratio. The torsional capacity of the tower with a prestress ratio of 0.6 is 127.67% higher than that of the tower with a prestress ratio of 0.1. However, when the prestress ratio exceeds 0.6, the torsional capacity decreases with the increase in prestress. The preliminary analysis shows that when the prestress is low, increasing the prestress can stabilize the boundary conditions at the joints, which enhances the tower's torsional capacity. However, when the prestress becomes too high, the concrete in the tower segment will quickly reach the ultimate compressive strain due to its large initial compression, leading to crush. This results in a

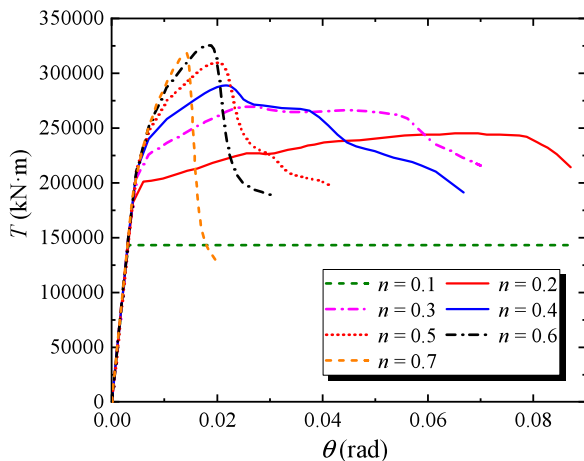


Fig. 23. T - θ curves of prestressed concrete towers with different prestress ratios.

decrease in the torsional capacity. Additionally, the torsional ductility of the tower decreases significantly as the prestress increases, especially when the prestress ratio exceeds 0.4. The preliminary analysis suggests that the torsional ultimate state shifts from steel reinforcement yielding to concrete compressive failure as the initial compression increases, resulting in a transition from ductile failure to brittle failure. It is worth noting that when the prestress ratio is 0.1, the tower exhibits the joint failure. After a brief initial stiffness stage, the torsional angle increases almost horizontally, indicating that the tower has not even reached the cracking load when the joint fails. The tower segment is still almost in the elastic stage, but due to the joint failure, the tower can no longer withstand a higher torque. This further proves that the prestress ratio significantly affects the torsional behavior of the tower. If the torsional capacity is satisfied, the prestress ratio should be minimized as much as possible, thereby providing good torsional ductility for the tower. The commonly used range of prestress ratios between 0.15 and 0.25 in engineering practice can meet this requirement.

4. Mechanism analysis

4.1. Typical load-displacement curves

Based on the analysis of the above parameters, it can be concluded that the prestressed concrete towers under different parameters exhibit similar torsional modes, which can be characterized by two typical T - θ curves: one representing torsional failure of the tower segment and the other representing joint failure, as summarized in Fig. 24. Preliminary analysis can be obtained based on the typical curves.

The typical T - θ curve of torsional failure of the tower segment can be divided into three stages by three characteristic points: elastic stage (Stage I), elastic-plastic stage (Stage II), and plastic development stage (Stage III). In Stage I, the torque and torsion angle are generally linearly related. After point A', the concrete begins to crack, and the stiffness of the T - θ curve slightly decreases. By the time it reaches point A, the concrete cracks significantly develop, marking the end of the elastic stage. In Stage II, a portion of the concrete ceases to participate in the load bearing, causing a noticeable decrease in stiffness and transitioning into the elastoplastic stage. After reaching point B', the stiffness further decreases. The fewer the steel reinforcements in the tower, the more noticeable the reduction in stiffness. After reaching point B, all the steel reinforcements in the critical section yield. At this point, the torque on the tower reaches the ultimate torque $T_{u,t}$. Thus, point B represents the ultimate state. In Stage III, the torque decreases slightly, and the torsional angle can continue to increase. Meanwhile, as the torsional angle grows, the compressive strain of the concrete increases continuously until it reaches the ultimate strain. After point C, the concrete can no longer withstand compressive stress induced by the torque and then undergoes crushing, resulting in rapid decreasing of the torsional capacity and brittle failure of the tower segment.

The typical T - θ curve of joint failure can be divided into two stages: torsional stage of tower segment (Stage I), slip stage of horizontal joint (Stage II). In stage I, in the initial loading stage, no joint slip occurs, and the torsional mechanism is identical to that of the torsional failure mode of the tower segment. This continues until the torque reaches the critical value that induces joint slip, at which point the curve reaches point A, marking the end of this stage. In Stage II, the torque experienced by the tower reaches the critical value that induces joint slip, which corresponds to the ultimate torque $T_{u,j}$ of joint failure. Thereafter, torsional slip occurs at the joint, leading to the failure of the tower's boundary conditions. The stress in the tower segment ceases to increase, and the torque remains nearly constant or slightly decreases, while the torsional angle continues to grow.

The above analysis only provided a preliminary prediction of the torsional process based on the T - θ curves obtained from parametric analysis. Thus, representative models were selected to further study the full-process torsional mechanism of the tower. The model labels and

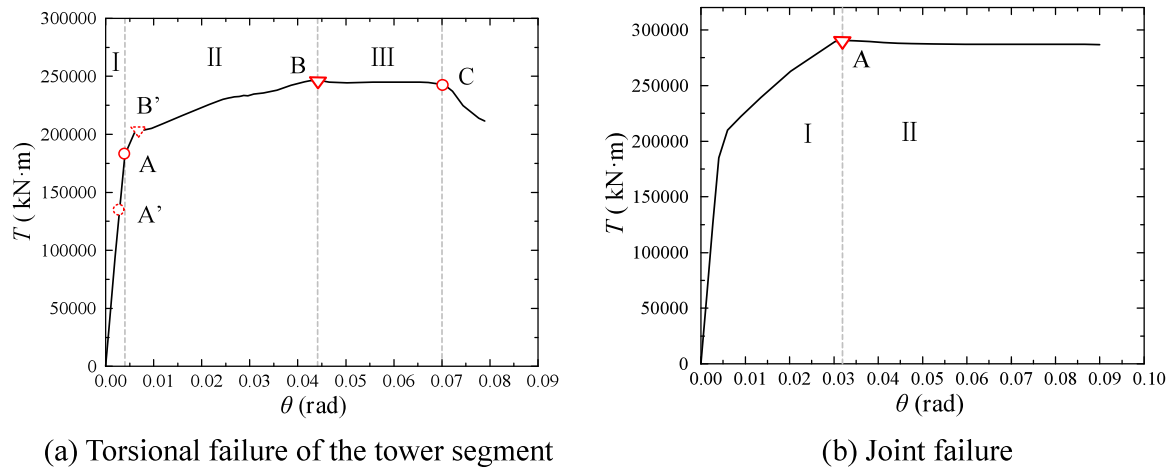


Fig. 24. Typical T - θ curves of prestressed concrete towers.

their corresponding parameters used in the analysis are listed in Table 3, while other unspecified parameters remain consistent with the basic model.

4.2. Stress states

4.2.1. Stress state of concrete

Typical models were selected to analyze the concrete stress states at characteristic points A and B. The magnitudes and directions of principal tensile stress in the concrete are shown in Fig. 25. Under the torsional shear stress, diagonal principal tensile stresses are formed in the tower. It can be found that the angle between the direction of the principal tensile stress and the axial direction of the tower increases with the prestress ratio. At point A, the tensile stress distribution along the tower height is relatively uniform, indicating that most areas of the tower experiences unrestrained torsion before point A. The tensile stress is slightly higher in the top region, which is attributed to the greater stiffness of the transition section. Consequently, initial cracks are most likely to appear in the top region of the tower, which is consistent with experimental observation. After point B, as concrete cracks propagate, tensile stress at the crack locations begins to decrease. Following stress redistribution, tensile stress increases in other regions, leading to a downward shift of high-stress zones. The critical cross-sections were selected within the high-stress region for examining the stress distribution across the section, as illustrated in the Fig. 25. At point A, the stress flow aligns closely with Saint-Venant's torsion theory, with shear stress forming a uniform circulation around the section. This indicates that before point A, the tower remains largely in the elastic torsion stage. Additionally, the shear flow in the cross-section is unevenly distributed

along the thickness direction, showing a trend of being smaller on the inside and larger on the outside. This indicates that the shear flow of the tower does not fully conform to Bredt's thin-walled shear stress distribution. However, for the tower T-d-2 shown in Fig. 25(c), the stress magnitudes of the inside and outside are relatively similar due to a large hollow ratio of 0.92. Conversely, for the tower T-t-2 with a smaller hollow ratio, the stress distribution along the thickness direction exhibits a clear gradient. This proves that the current engineering design approach of broadly applying Bredt's shear stress distribution to calculate the shear stress of the cross-section is not sufficiently accurate.

To more clearly compare the sectional stress distribution of towers with different parameters, the concrete tensile stress σ_{ct} at critical cross-sections were extracted, as shown in Fig. 26. It can be observed that at point A, the tensile stress distribution is uniform for all towers. However, at point B, due to the redistribution of stresses after concrete cracking, the section stresses no longer conform to Saint-Venant's torsional stress distribution. The tensile stress values of both the inside and outside fluctuate, showing alternating peaks and troughs along the circumferential direction of the section. The troughs represent the areas where the concrete has cracked, and the tensile stress approaches zero when the crack is severe. On the other hand, the peaks correspond to the concrete between cracks, which can still bear tensile stress well, though the stress is typically lower than the tensile strength. At point B, the stress difference between the inside and outside becomes more pronounced compared to point A. The stress difference is greater when the hollow ratio is smaller. Tower T-t-2 with a thickness of 600 mm exhibits severe stress fluctuations on the outside. However, the stress in the middle and inside of the section remains uniform and at a higher level. The outside has experienced tensile softening, but the inside is still able to resist the

Table 3
Parameters of mechanism analysis.

Model labels	Diameter D (mm)	Thickness t (mm)	Prestress ratio n	Horizontal reinforcement ratio ρ_h	Vertical reinforcement ratio ρ_v
T-0	7000	400	0.2	0.99 %	1.31 %
T-t-1	7000	200	0.2	0.99 %	1.31 %
T-t-2	7000	600	0.2	0.99 %	1.31 %
T-t-3	7000	600	0.14	0.99 %	1.31 %
T-D-1	4000	400	0.2	0.99 %	1.31 %
T-D-2	10,000	400	0.2	0.99 %	1.31 %
T-D-3	10,000	400	0.14	0.99 %	1.31 %
T-n-1	7000	400	0.1	0.99 %	1.31 %
T-n-2	7000	400	0.4	0.99 %	1.31 %
T-n-3	7000	400	0.6	0.99 %	1.31 %
T-sh-1	7000	400	0.2	0.34 %	1.31 %
T-sh-2	7000	400	0.2	1.62 %	1.31 %
T-sv-1	7000	400	0.2	0.99 %	0.45 %
T-sv-2	7000	400	0.2	0.99 %	2.18 %

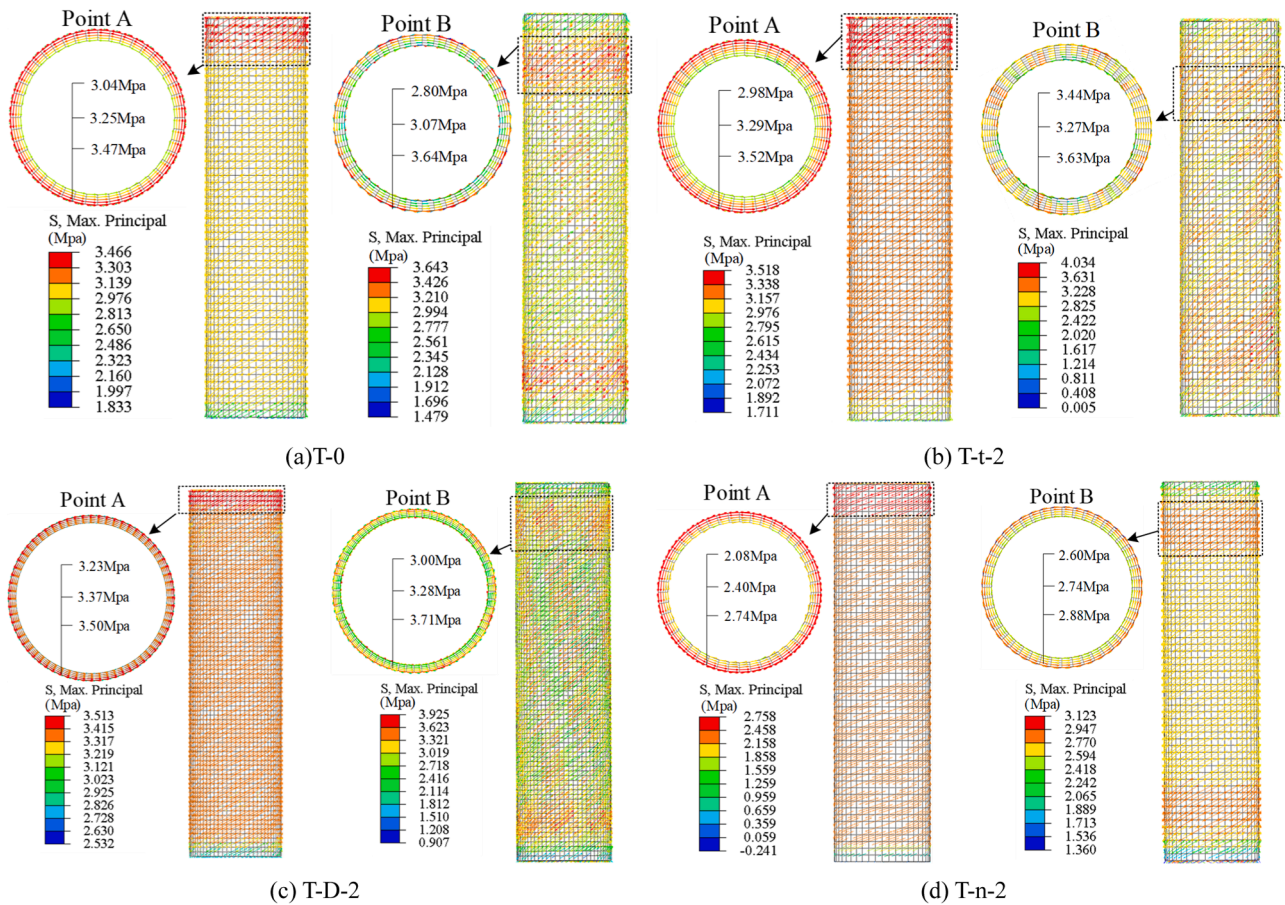


Fig. 25. Principal tensile stress of concrete.

torque due to the redistribution of stress. This further demonstrates that, for the tower with small hollow ratio, when the middle and inside of the section have not yet reached the average shear stress, the outside may have already suffered severe cracking, and the torsional capacity may have begun to decline. In such cases, using the Bredt's shear stress distribution for analysis and calculation would be unconservative. The tower with a low reinforcement ratio T-sh-1 shows significant fluctuations in the stress distribution on both the inside and outside, with several stress troughs approaching zero and stress peaks reaching the tensile strength. This indicates that the reinforcement is unable to effectively transfer the tensile stresses from concrete cracking, leading to a significant difference from the stress distribution before cracking. In contrast, the tower T-D-2 with a diameter of 10,000 mm allows the concrete to better resist the torque due to its larger torsional stiffness. The stress distribution is more uniform, and the stresses on the inside and outside are also closer to each other due to the higher hollow ratio.

The tensile stress distribution in the tower T-n-3 with larger prestress exhibits a different pattern, as shown in Fig. 26(f). Due to the large initial compressive stress in the concrete, the overall tensile stress remains low after torsion. At point A, the difference in tensile stress between the inside and outside is considerable. The higher the prestress ratio, the lower the tensile stress, and the more pronounced the stress difference between the inside and outside. At point B, the stress difference between the inside and outside decreases, and the tensile stress increases slightly but remains relatively low overall. This indicates that, the principal tensile stress induced by torque gradually offsets the initial compressive stress. However, it is significantly lower than the tensile strength. The large prestress effectively delays the development of concrete cracks, resulting in a relatively uniform stress distribution across the section. In summary, the concrete between the cracks can still

resist the tensile stress induced by torsion at the ultimate state. However, when the axial compression ratio is excessively high, the tower cannot fully utilize the tensile strength.

Towers with large prestress exhibit relatively low tensile stress even at the ultimate state. Combined with the brittle failure observed in the $T-\theta$ curves from the parametric analysis, this indicates that their ultimate state might be governed by concrete compressive failure. The compressive stress of the concrete at the ultimate state is shown in Fig. 27. It can be observed that the tower with a prestress ratio of 0.2 remains in the elastic compressive stage at the ultimate state, with relatively low compressive stress. In contrast, the tower with a prestress ratio of 0.6 exhibits significantly higher compressive stress, with a significant diagonal compression strip on the outside of the tower. The orientation of the strip is opposite to that of the tower with a prestress ratio of 0.2. This is because, as the initial compressive stress increases, the angle between the principal compressive stress direction and the tower's axis direction gradually decreases, causing the orientation of the high-stress compression strip to shift. It can be observed that, although the failure mode of the tower with a prestress ratio of 0.6 is governed by concrete compressive failure, the peak stress does not exceed the uniaxial compressive strength. This can be attributed to that the concrete is subjected to a combined tensile-compressive stress state under torsion, where the compressive strength is reduced compared to its uniaxial strength.

4.2.1. Stress state of steel reinforcement

The stress of steel reinforcement σ_s along the cross-section at the ultimate state are shown in Fig. 28. In the figure, H represents horizontal reinforcement, and V represents vertical reinforcement. From the basic model T-0, it can be observed that the stress of horizontal reinforcement

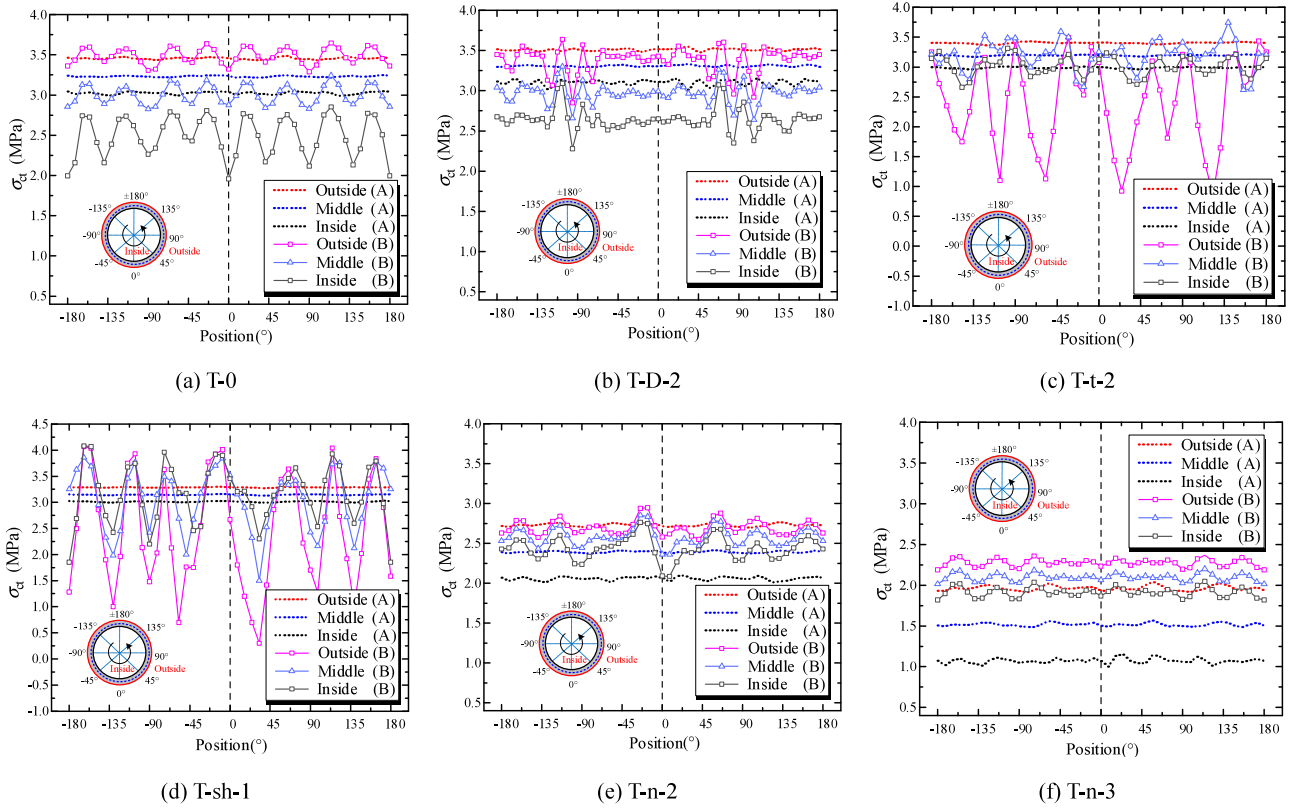


Fig. 26. Concrete tensile stress distribution along the cross-section in the circumferential direction.

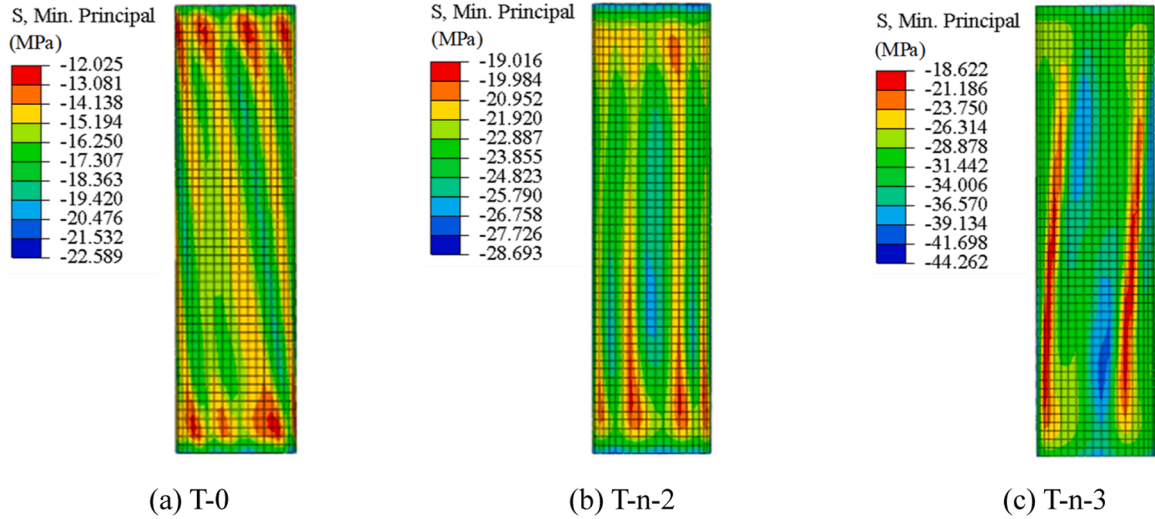


Fig. 27. The compressive stress state of the concrete at ultimate state.

is high, with most of the outer reinforcement reaching the yield strength, effectively contributing to the torsional resistance. The stress distribution exhibits slight fluctuations, which is consistent with the tensile stress distribution in the concrete. The reinforcement stress is higher at the locations of concrete cracks and slightly lower between the cracks. In contrast, the stress of the inner reinforcement is lower, with a smaller yield region compared to the outer layer. The stress fluctuations are also smaller. For the tower T-t-2 with a lower hollow ratio, the stress of the inner and outer reinforcement shows a greater difference, with the inner reinforcement almost not yielding. Only when the tower's hollow ratio is relatively large, the stress of the inner and outer layer can become similar, such as the tower T-D-2 with a diameter of 10,000 mm. This

again illustrates that the stress distribution of the tower cannot well satisfy the Bredt's shear stress distribution. Due to the differences in stress between the inside and outside of the section, the inner and outer reinforcement will contribute to the torsional resistance to different extents. The outer reinforcement usually yields before the inner reinforcement, so the effects of inner and outer reinforcement should be considered separately.

For the vertical reinforcement, both the inner and outer layers exhibit lower stress, well below the yield strength. This is attributed to that the prestressing force causes an initial compressive stress to the vertical reinforcement. After torsion applied, the stress state of vertical reinforcement gradually transitions from compression to tension.

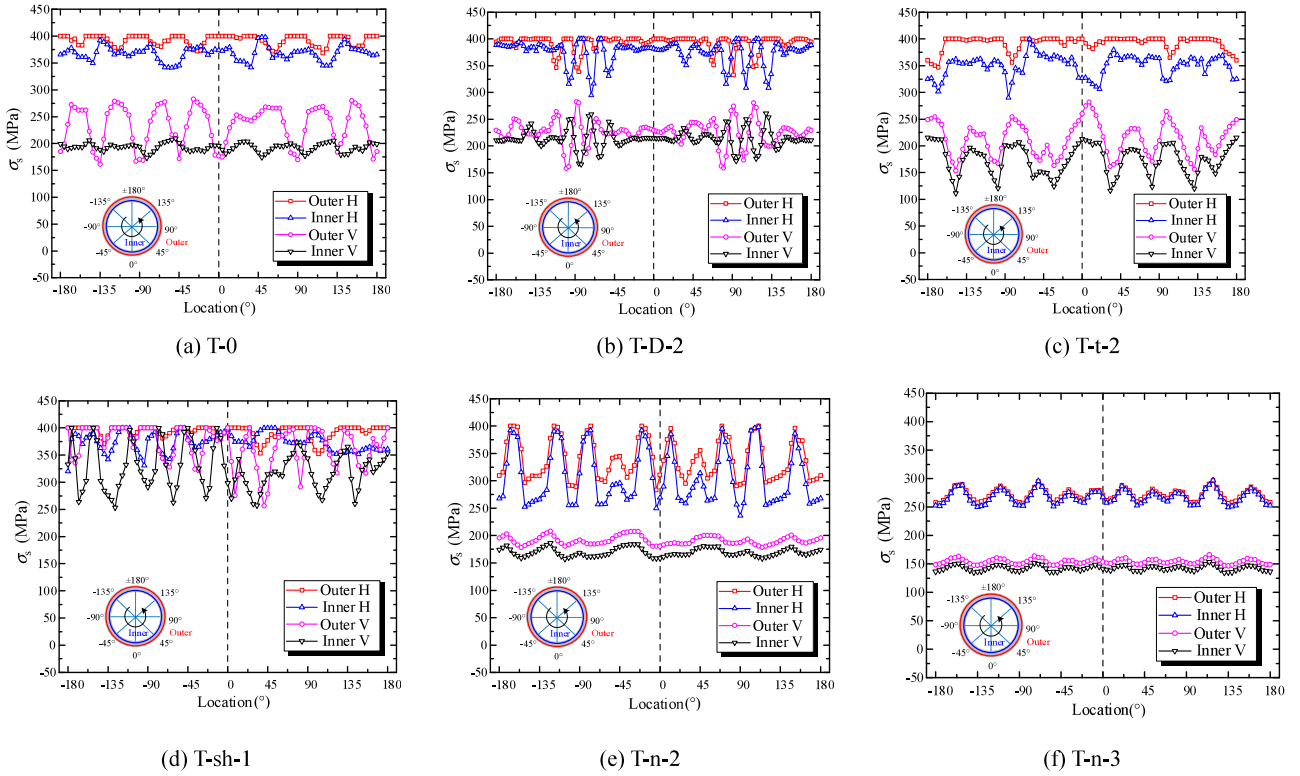


Fig. 28. Stress distribution of reinforcement along the cross-section at the ultimate state.

Therefore, when the horizontal reinforcement yields in tension, the stress in the vertical reinforcement remains relatively low, and the steel strength is not fully utilized. This phenomenon is more significant as the prestress ratio increases.

The tower T-n-3 with a prestress ratio of 0.6 shows that the horizontal reinforcement has not yet yielded, and the stress is far below the yield strength. As the prestress ratio increases, the stress in the reinforcement further decreases, and the development of the strength utilization is further weakened. However, the higher prestress delays the development of concrete cracks, resulting in a more uniform and closer distribution of stress between the inner and outer reinforcement. Combined with the concrete compressive stress state shown in Fig. 27, it further proves that excessive prestress will lead to concrete compressive brittle failure, where the reinforcement has not yielded, making it difficult to utilize the ductility of the steel reinforcement.

In contrast, the tower T-sh-1 suffers from severe concrete cracking on both inside and outside due to the low reinforcement ratio. The tensile stresses in the concrete are quickly transferred to the reinforcement, resulting in higher and more uniform stress distribution in the inner and outer reinforcement. A large area of reinforcement yields, and even some vertical reinforcement reaches the yield strength. While the strength utilization of the reinforcement is relatively high, the tower with low reinforcement ratio experiences rapid failure once concrete cracking occurs. Combining this with the $T-\theta$ curve, it can be observed that its torsional capacity and ductility are both low, indicating a tensile brittle failure mode.

4.3. Load distribution

To further analyze the contributions of concrete and reinforcement to the torsional capacity of the tower, the torque values and proportions of concrete and reinforcement at the ultimate state (point B) were shown in Fig. 29. It can be observed that the torsional capacity of the tower T-D-2 increased by 116.98 % compared to the tower T-0, with the torque proportion of the concrete increasing by 0.55 %. In contrast, the

torsional capacity of the tower T-t-2 only increased by 23.17 % compared to the tower T-0, but the torque proportion of the concrete increased by 9.13 %. This indicates that, when the reinforcement ratio is unchanged, the torque carried by the concrete increases as the tower dimension grows, and the increase is more pronounced when the tower thickness is increased. This is because increasing the thickness moves the inner reinforcement towards the center of the cross-section, reducing the torque carried by the inner reinforcement and thereby increasing the torque proportion of the concrete. Additionally, increasing the diameter increases the radius of the reinforcement framework, which simultaneously enhances the torsional contribution of both the reinforcement and the concrete, leading to a more significant improvement in the torsional capacity of the tower. On the other hand, increasing the reinforcement ratio will result in an increase in both the torsional capacity of the tower and the torque proportion of the reinforcement. Besides, the torque proportion of the concrete is decreased. This indicates that the increase in torsional capacity resulting from a higher reinforcement ratio primarily comes from the contribution of the steel reinforcement itself in resisting torsion, while the contribution of the concrete is less affected by variation in the reinforcement ratio.

It can be observed that as the prestress ratio increases, the torque borne by the concrete significantly increases, while the torque borne by the reinforcement decrease. The initial compressive stress provided by prestress delays the cracking of the concrete, allowing it to better resist torque. However, an excessively large prestress ratio leads to compressive failure of concrete, preventing the reinforcement from effectively utilizing its strength, thereby significantly reducing the torque proportion of the reinforcement. In contrast, although the prestress ratio of the tower T-n-1 is only 0.1, the torque proportion of the concrete reaches as high as 81.29 %. This is because the low prestress results in a joint failure. At this point of failure, the tower remains in the elastic stage, and the stress of the reinforcement is still low, leading to a smaller torque borne by the reinforcement. This trend is also evident in the thickness and diameter series. The contrast is more pronounced in the towers T-D-2 and T-D-3, both with a diameter of 10,000 mm but different prestress

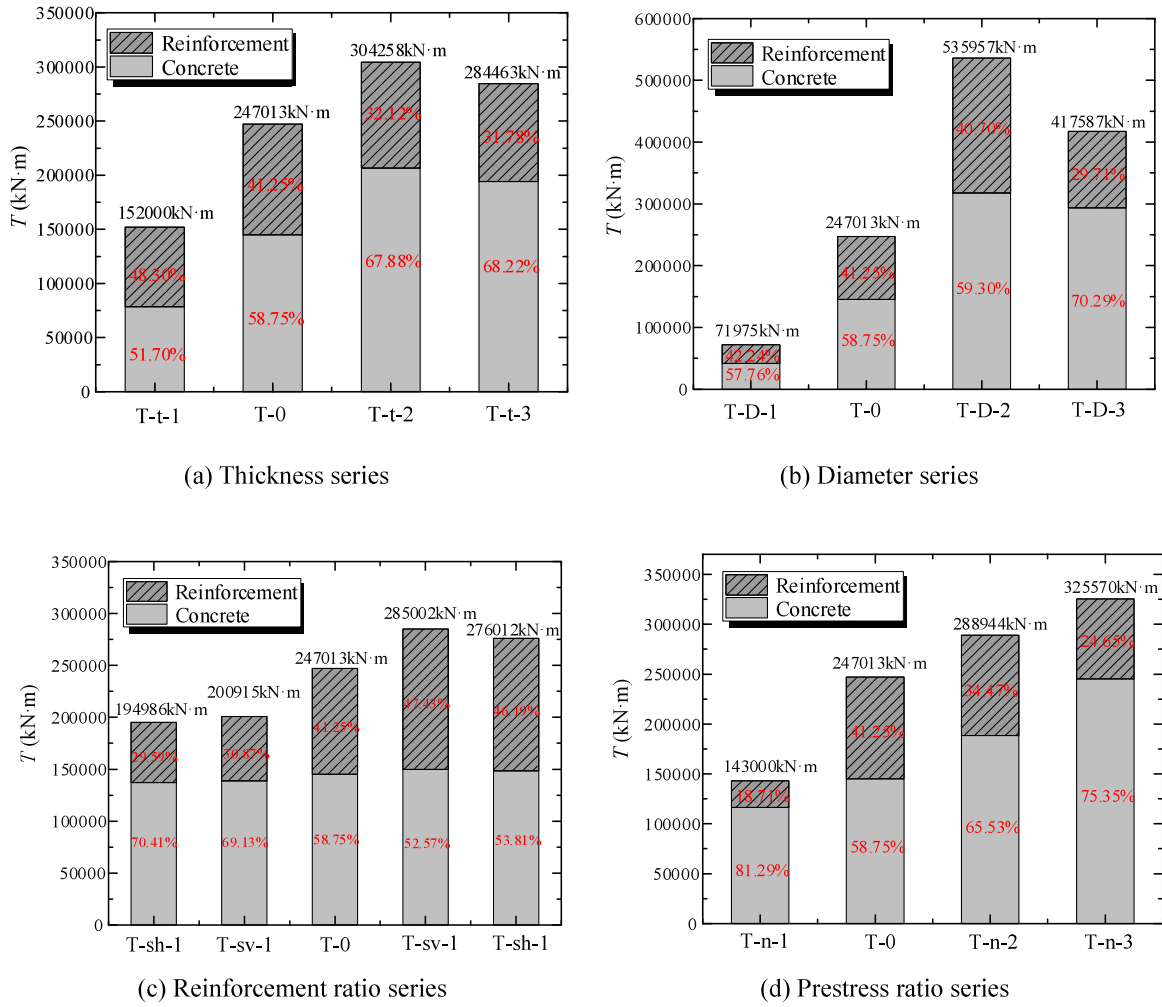


Fig. 29. Torque proportions of concrete and reinforcement.

ratios. In the tower T-D-3, the torsional capacity of the tower segment far exceeds the critical torque for joint failure, resulting in earlier joint failure and lower reinforcement stress. Consequently, the torque proportion of the reinforcement is smaller.

4.4. Shear force of joint interface

To further investigate the critical state of joint failure, the interface shear forces of different towers at point A and point B are shown in Fig. 30. Based on Coulomb's friction theory, the maximum static friction force $F_{f,max}$ was calculated as following equation:

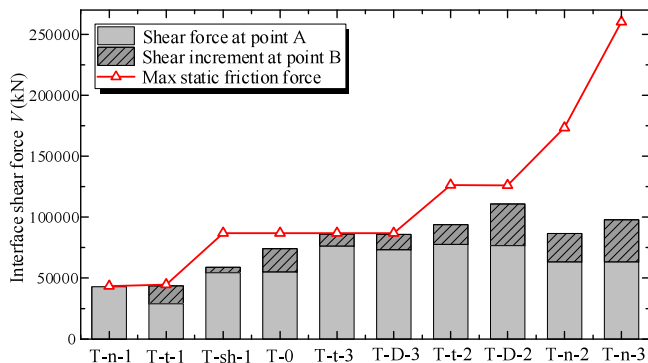


Fig. 30. Interface shear force of horizontal joint.

$$F_{f,max} = \mu(P + N) \quad (5)$$

Where, P is the prestressing force; N is the vertical load; μ is the frictional factor of the joint interface.

According to Eq. (5), the maximum static friction force depends only on the total compression applied to the interface. It can be observed from Fig. 30 that the interface shear force increases with the increase in tower dimension and reinforcement ratio. According to the static equilibrium theory, the interface shear force should be equal to the shear force generated by the torque. Therefore, before the boundary condition of the joint fails, the interface shear force increases as the torque increases. The torque at point A corresponds to the cracking torque and the torque at point B corresponds to the torsional capacity. The difference between the cracking torque and the torsional capacity determines the variation in interface shear force at these two characteristic points. However, as observed from the shear forces of T-n-1, T-t-1, T-t-3, and T-D-3, regardless of the tower dimensions, the shear forces do not exceed the maximum static friction force $F_{f,max}$. This can be explained by the sliding friction law: the joint will begin to rotate after the torsional shear force on the interface exceeds the maximum static friction force, and the shear force no longer increases but remains constant, equal to the sliding friction force. For towers exhibiting the joint failure mode, the interface shear force at point B is always slightly lower than the maximum static friction force $F_{f,max}$.

The above analysis explains the reasons for the differences in interface shear forces between towers T-t-2 and T-t-3, as well as T-D-2 and T-

D-3, which have identical dimensions and reinforcement ratios. Due to the greater maximum static friction force generated by the prestress, the interface shear force of T-t-2 is greater than that of T-t-3, and the interface shear force of T-D-2 is greater than that of T-D-3. In combination with different failure modes, it can be concluded that when the tower undergoes torsional failure, the ultimate shear force is controlled by the torsional capacity of the tower. When joint failure occurs, the ultimate shear force is controlled by the maximum static friction force at the joint interface. From the interface shear force of T-n-2 and T-n-3, it can be observed that the greater the prestress, the larger the maximum static friction force generated at the interface, resulting in a greater difference between the ultimate shear force and the maximum static friction force. Therefore, the higher the prestress ratio, the less likely the tower is to experience joint failure, allowing the torsional capacity of the tower segment to be better utilized.

4.5. Prestress variation

To investigate the prestress variation process of towers, the prestress (σ_p)-torsional angle (θ) curves of towers are shown in the Fig. 31, illustrating the stress variation of four prestressed tendons located at key positions (No.1~No.13, as shown in Fig. 31). Before point A, the

prestress remains almost unchanged, indicating that the deformation of the tower is minimal before concrete cracking. After point A, the diagonal cracks in the tower segment develop continuously, causing the concrete between the cracks to be compressed, resulting in compressive deformation of the tower segment and a reduction in prestress. Most of the towers experience a reduction in prestress to its minimum value near point B, after which it turns to increase. This indicates that the compressive deformation of the tower continues until the ultimate state. Once the concrete cracks have developed uniformly, the cracked concrete must rotate along the inclined crack surface in a spiral direction around the tower axis, causing the tower segment to elongate. As a result, the prestressed tendons also elongate. During the torsion process after point B, uneven deformations occur at various locations due to the failure of the tower, leading to progressively difference in the prestress variation of the steel tendons at different positions.

The following towers do not fully conform to the above-mentioned variation pattern: (1) The tower T-t-1 experiences joint failure before the torsional failure of the tower segment. During the slip of the joint, the deformation of the tower segment is uniform, so the prestress variation is small. (2) The tower T-sh-1 with small reinforcement ratio, exhibits a brittle failure mode. Its characteristic points A and B are very close, and the prestress variation is small. However, due to the sudden

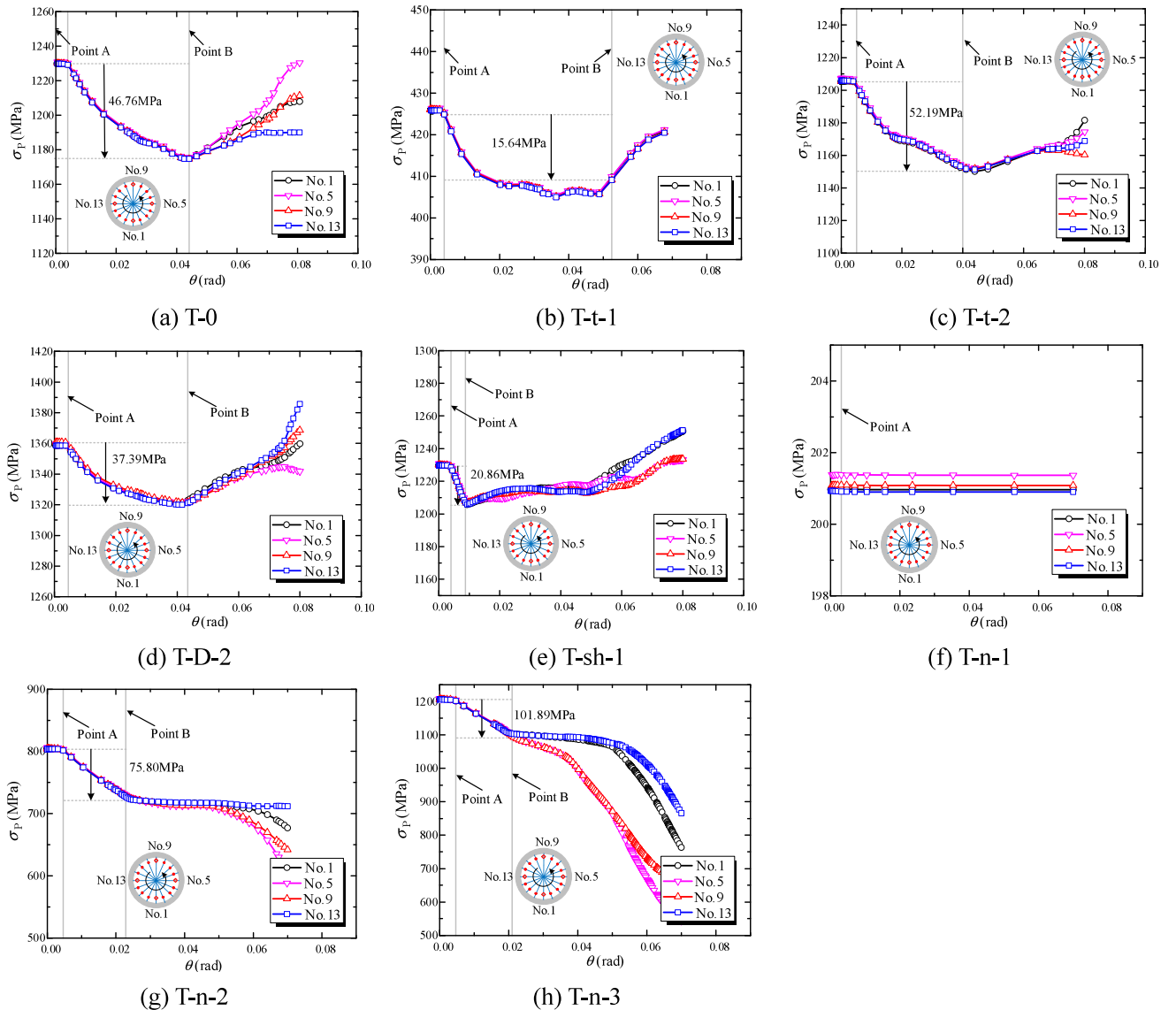


Fig. 31. Prestress (σ_p)-torsional angle (θ) curves.

failure of the tower segment, the prestress decrease rate is significantly larger. (3) The tower T-n-1, with a small prestress ratio, reaches its ultimate capacity due to joint failure while the tower segment is still in the elastic stage. At this point, the tower segment has not undergone deformation, so the prestress remains almost constant. (4) The tower T-n-2 and T-n-3 with high prestress ratios, do not experience complete development of concrete cracks when reaching the ultimate state. As a result, after point B, the tower segment continues to be compressed, causing the prestress to further decrease. This phenomenon is more pronounced as the prestress ratio increases.

To clarify the magnitude of the prestressing force on the tower at the ultimate state, the initial prestressing force and the total prestressing force at the ultimate state were compared, as shown in Fig. 32. At the ultimate state, the total prestress of each tower shows varying degrees of reduction. The reduction is minimally influenced by the tower dimensions and reinforcement ratios but is significantly affected by the type of failure mode and the magnitude of the initial prestress. The prestress reduction is relatively small for towers exhibiting joint failure modes. For instance, the tower T-n-1, which experiences joint failure in the elastic stage, shows a prestress reduction of only 0.01 %. In contrast, the tower T-n-3, with a higher initial prestress, undergoes compressive failure, leading to significant compression-torsion deformation of the tower segment and a prestress reduction of 9.37 %. For other towers with a prestress ratio of 0.2 (a commonly used prestress ratio), the maximum prestress reduction rate is only 3.55 %. Overall, for towers with reasonably designed prestress ratio that experience ductile torsional failure of the tower segment, the prestress reduction is minimal. This suggests that the loss factor of 0.05 specified in the IEC 61400-6-2020 [57], which accounts for reductions in external unbonded prestress, is sufficient to cover the prestress reduction at the ultimate state for towers exhibiting ductile torsional failure mode. Therefore, a conservative loss factor of 5 % is adopted in subsequent analyses to calculate the prestress at the ultimate state. For towers exhibiting joint failure mode, the prestress variation is typically minimal before joint slips, allowing the impact of prestress reduction to be neglected.

4.6. Full-process torsional mechanism

Based on the above analysis results from the previous sections, the full-process torsional mechanism of prestressed concrete towers can be summarized.

Under the prestressing force, the initial compressive stress exists in the tower, and the shear flow is generated on the section after the tower is subjected to torque. Considering a concrete element, under the combined effects of torsional shear stress and initial compressive stress, principal tensile stress σ_{tp} and principal compressive stress σ_{cp} are produced in the α direction, as shown in Fig. 33. The angle α between the principal compressive stress direction and the tower axis is $<45^\circ$. Before the principal tensile stress exceeds the tensile strength, the torsional

performance of the tower generally conforms to Saint-Venant's elastic torsion theory, with the maximum shear stress occurring at the outer edge of the section. After the principal tensile stress exceeds the tensile strength and the torque is larger than the cracking torque, the concrete will crack. The crack angle with respect to the tower axis is less than 45° , and the greater the prestress, the smaller the crack inclination angle. At this stage, the stress borne by the reinforcement remains relatively small. The cracking torque is mainly influenced by the dimensions and concrete strength of the tower segment, while the impact of the reinforcement is negligible.

After cracking, the tensile stress at the crack is released and transferred to the reinforcement, causing a rapid increase in reinforcement stress. Meanwhile, the reinforcement stress between the cracks remains low, and the concrete in this region can continue to bear the tensile stress until it exceeds the tensile strength, at which point the concrete here enters a cracked state again. As the torque increases, cracks appear successively according to this pattern and develop along the direction of the principal compressive stress, until all the cracks are formed. The concrete is then uniformly divided into multiple oblique strips by the diagonal cracks, which are spirally distributed around the tower segment, as shown in Fig. 33. The concrete oblique strips are compressed along the direction of the cracks and tensile along the direction perpendicular to the cracks. The concrete between the cracks can still bear tensile stress, but the tensile stress is slightly smaller than the tensile strength, so new cracks cannot form. Subsequently, the increase in tensile stress due to the torque is mainly borne by the steel reinforcement. The vertical reinforcement gradually transitions from a compressive state to a tensile state, while the tensile stress in the horizontal reinforcement continues to grow. The stress of the outer reinforcement is significantly higher than that of the inner reinforcement, so the outer reinforcement generally yields first. For towers with appropriate reinforcement ratios, the tower will reach the ultimate state after the reinforcement yields. The concrete with spiral cracks and the reinforcement form a new cooperative torsional system, with both being able to participate in tension. At the ultimate state, the prestress decreases, but the reduction does not exceed 5 % generally.

After the ultimate state, as the plasticity of the reinforcement develops, the torsional angle can continue to increase, but the torque remains almost constant. During the plastic development, the compressive stress in the concrete continues to increase. Since the concrete diagonal strips are in a combined tensile-compressive stress state, their compressive strength is significantly lower than the concrete's uniaxial compressive strength, which will quickly lead to the concrete crushing, resulting in a rapid decline in torsional capacity. When the initial prestress is high, the concrete diagonal strips may fail due to compression before the reinforcement yields, resulting in weaker torsional deformation and exhibiting typical brittle failure mode. This should be avoided in design. Therefore, even though the ultimate capacity at the point of concrete crushing may be higher than that at the point of reinforcement yielding, the yielding of the reinforcement should be considered as the ultimate state for the tower. Hence, the initial prestress of the tower should be limited, and based on the aforementioned analysis, the total axial compression ratio m should be restricted to below 0.4 to avoid compressive failure of the tower segment.

The above describes the complete mechanism for torsional failure of the tower segment, whose premise is that the torque does not exceed the ultimate torque of the joint interface $T_{u,j}$. Otherwise, joint failure will occur. Under torsion, the joint interface experiences torsional shear forces, causing the tower segment to tend to rotate in the direction of torsion, as illustrated in Fig. 34. However, this shear force can be counteracted by the frictional force provided at the joint interface, known as the torsional reaction force T_j . The value of the torsional reaction force equals the torque applied to the tower and increases as the torque increases. Until the torsional shear force exceeds the maximum static friction force at the joint interface, joint failure occurs. Afterward, the torque remains nearly constant, and the tower becomes a movable

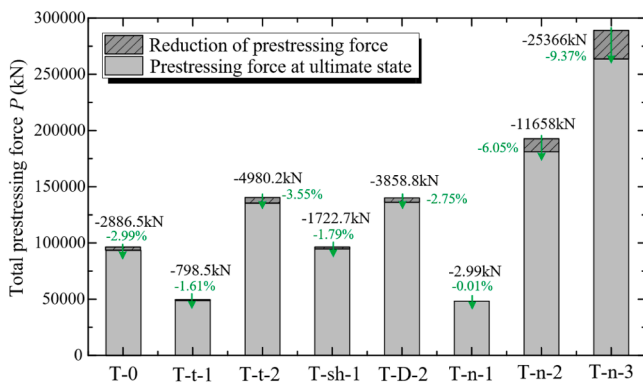


Fig. 32. Variation of prestressing force at ultimate state.

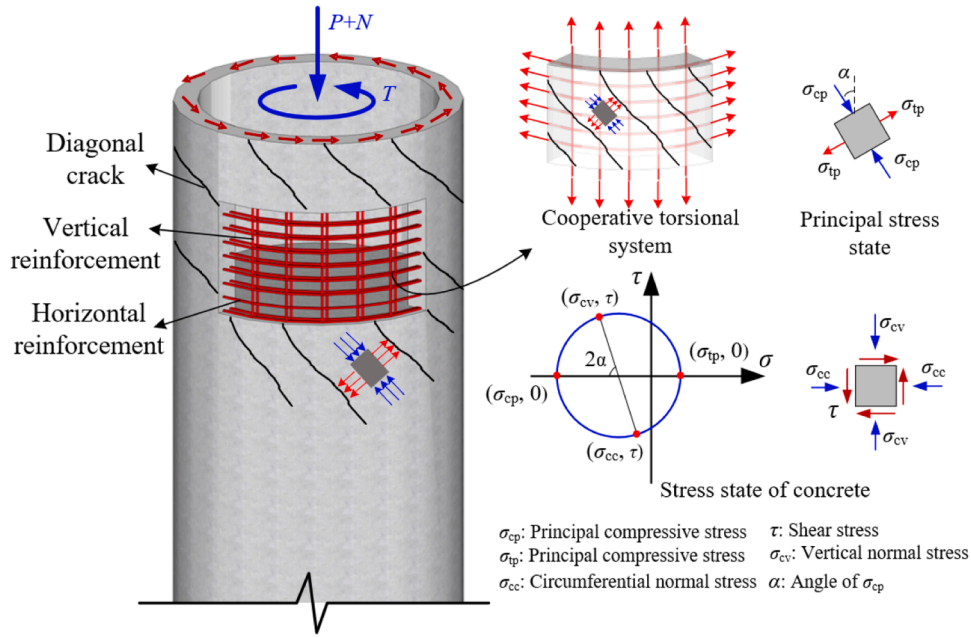


Fig. 33. Stress state of the prestressed concrete tower.

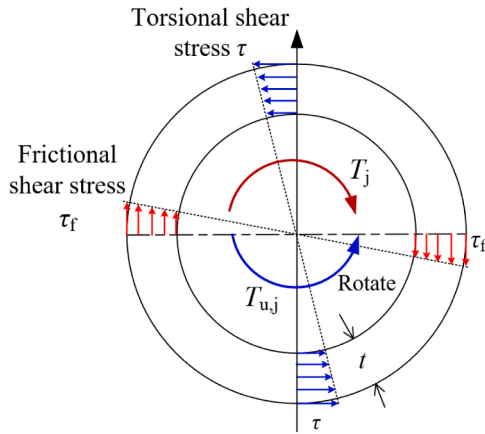


Fig. 34. Mechanism of joint failure.

mechanism capable of rotating along the joint interface. The occurrence of the joint failure prevents further damage to the tower segment. At this point, the ultimate torsional capacity of the tower is governed by the ultimate torque $T_{u,j}$ corresponding to the joint failure. The increase in prestress will enhance the frictional force, thereby increasing the ultimate torque $T_{u,j}$. Different values of $T_{u,j}$ lead to varying stress responses in the tower segment before joint failure. When $T_{u,j}$ is sufficiently low, the tower segment may remain in the elastic stage, or even without concrete cracking, the tower could lose its load-bearing capacity due to joint failure.

5. Development of calculation method of ultimate torsional capacity

5.1. Torsional capacity of tower segment

Based on the mechanistic analysis, when the tower undergoes the torsional failure of the tower segment, the ultimate capacity is governed by the tower segment's torsional capacity $T_{u,t}$. For towers with appropriate reinforcement ratios, concrete and reinforcement can form a cooperative torsional system, while prestressed steel tendons do not bear

the torque. The reinforcement ratio has a minor impact on the torque borne by the concrete but significantly affects the torque borne by the reinforcement itself. Therefore, referring to the method of linearly superimposing the bearing capacities of concrete and reinforcement in the GB50010–2010 [56], the contributions of concrete and reinforcement to the torsional capacity $T_{u,t}$ are considered separately:

$$T_{u,t} = T_{us} + T_{uc} \quad (6)$$

Where, T_{us} is the torsional capacity of the reinforcement; T_{uc} is the torsional capacity of the concrete.

5.1.1. Torsional capacity of the reinforcement T_{us}

According to the mechanism analysis, the diagonal compression strip is formed after concrete cracking, which is consistent with the assumptions of the classical variable-angle truss model [14]. Therefore, the ultimate capacity of the reinforcement in the tower can be derived based on this model. In the traditional variable-angle truss model, it is assumed that the central portion of the cross-section carries minimal torsional shear stress. Thus, it hypothetically removes the central concrete far from the reinforcement, forming an assumed thin-walled hollow section, which can be analyzed using thin-walled torsion theory. For prestressed concrete towers, their inherently large hollow ratio eliminates the need for further removal of the cross-section. Additionally, the inner and outer layers of reinforcement positioned near the inner and outer walls of the tower, respectively. As revealed by the mechanism analysis, the shear stress does not always conform well to the Bredt's shear flow distribution. Therefore, the inner and outer reinforcements will exhibit different contributions to torsional capacity, which should be considered separately. It is not reasonable to broadly classify the cross-section as a single thin-walled hollow section based on the traditional variable-angle truss model. To accurately reflect the distribution of torsional shear stress, the middle line of the section is used as a boundary. The regions containing the inner and outer reinforcements are divided into two hollow thin-walled sections: the inner layer and outer layer of the tower, each with a thickness of $t/2$. The inner and outer reinforcements are considered to bear the torque of their respective regions, as illustrated in Fig. 35. According to Saint-Venant's torsion theory, the shear flow on the section is the integral of shear stress along the thickness direction of the section. The shear flow q_{out} of the outer

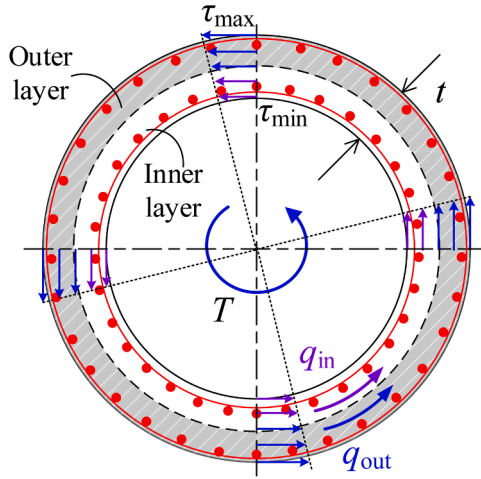


Fig. 35. Outer and inner layer of the tower section.

layer of the tower can be expressed as:

$$q_{out} = \int_{D_0}^D \frac{T}{I_p} \rho d\rho = \frac{Tt}{8I_p} (2D - t) \quad (7)$$

Where, D_0 is the middle diameter, $D_0 = D - t$; I_p is the polar moment of inertia of the cross-section.

According to the number of vertical reinforcements n_v , the outer layer of the tower section is considered as a variable-angle space truss composed of n_v plane trusses. Each truss model consists of $1/n_v$ of the concrete section along with its vertical and horizontal reinforcement. The height of each truss is $1/n_v$ of the perimeter of the tower section, as shown in Fig. 36. In this calculation model, the concrete is divided into several diagonal compression struts by cracks, which are subjected to compression only along the oblique direction. The vertical reinforcement within each truss model is regarded as the upper and lower chords, while the horizontal reinforcement is considered as the webs of the truss. Both the chords and webs are subjected to tension. The inclination angle of the compression strut is the angle of the concrete cracks, denoted as α . The total compression F_d born by the strut is caused by the shear flow q_{out} on the outer layer of the section:

$$F_d = \frac{q_{out} \pi D_{out}}{\sin \alpha} \cdot \frac{Tt(2D - t)\pi D_{out}}{8n_v I_p \sin \alpha} \quad (8)$$

Where, D_{out} is the diameter at the location of the outer reinforcement, which is the tower diameter minus the concrete cover on both sides.

F_d will cause compressive stress in the diagonal strut and generate tension in the upper and lower chords (vertical reinforcement). The average compressive stress σ_{ct} in the diagonal strut and the tension F_{v1} in each chord are calculated according to Eqs. (9) and (10), respectively.

$$\sigma_{ct} = \frac{F_d}{\frac{\pi D_{out}}{n_v} \cdot \cos \alpha \cdot \frac{t}{2}} = \frac{T(2D - t)}{4I_p \sin \alpha \cos \alpha} \quad (9)$$

$$F_{v1} = \frac{1}{2} F_d \cos \alpha = \frac{Tt(2D - t)\pi D_{out} \cdot \cot \alpha}{16n_v I_p} \quad (10)$$

Since all plane trusses are identical, the chord tensions F_{v2} to F_{vn} in the other trusses are equal to F_{v1} . The total compression F_d acting on the diagonal strut will cause a tension in the webs (horizontal reinforcement). In the calculation model, one horizontal reinforcement is assumed to be a web, and the length of the truss in the model is converted based on the horizontal reinforcement spacing s_h :

$$F_h = \frac{F_d \sin \alpha}{\frac{\pi D_{out}}{n_v} \cot \alpha} \cdot s_h = \frac{Tt(2D - t)}{8I_p} \cdot s_h \cdot \tan \alpha \quad (11)$$

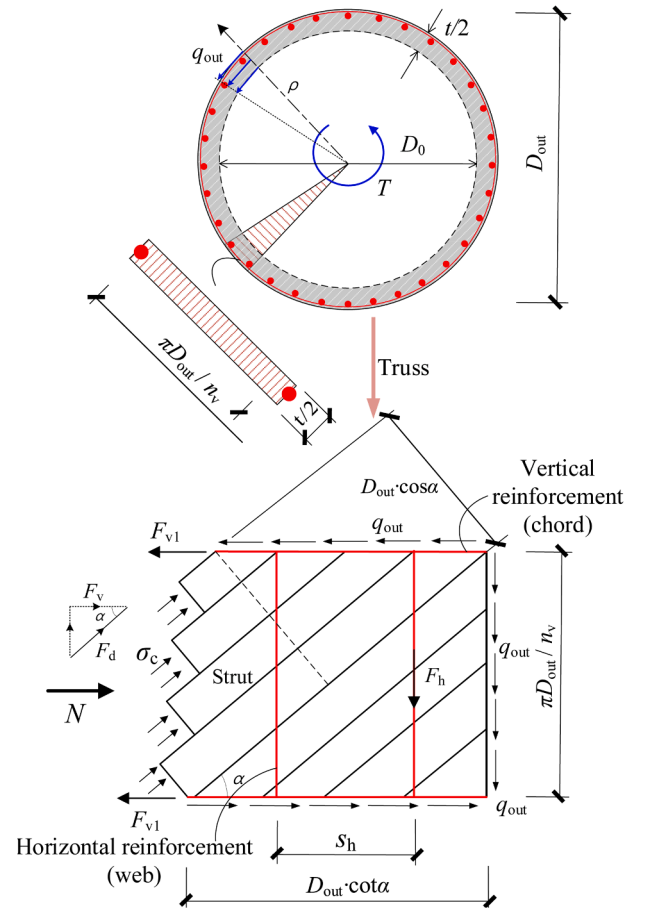


Fig. 36. Variable-angle truss model of the outer layer.

According to the variable-angle truss theory, for a tower with a reasonable reinforcement ratio, both the vertical and horizontal reinforcement will have yielded when the tower reaches its ultimate capacity. It should be noted that under the prestressing force and vertical load, the reinforcement has initial stress before being subjected to torsion. The initial stress in the vertical reinforcement is compressive, and the reinforcement gradually transitions from a compressive state to a tensile state under torsion. Thus, the total tensile force $F_{v,all}$ that borne by the vertical reinforcement in the outer layer of the tower can be expressed as:

$$F_{v,all} = f_{yv} A_{sv,out} + \frac{N_s}{2} \quad (12)$$

$$\text{With, } N_s = \frac{N_t}{E_c A_c + E_s A_{sv}} - E_s A_{sv}$$

Where, $A_{sv,out}$ is the sectional area of out layer reinforcement; N_s is the axial force borne by the vertical reinforcement; A_c is the sectional area of concrete; N_t is the total axial force, which is the sum of the prestressing force and the vertical load.

When the vertical reinforcement satisfies Eq. (10), the sum of the tensile forces of all the chords in the n_v trusses in the truss model should be equal to the total tensile force $F_{v,all}$ of the vertical reinforcement:

$$F_{v,all} = \sum_{i=1}^{n_v} (F_{vi} + F_{vi+1}) = \frac{Tt(2D - t)\pi D_{out} \cot \alpha}{8I_p} \quad (13)$$

By setting Eq. (12) equal to Eq. (13), the torsional capacity $T_{u,out}$ of the tower at the yielding state of the out layer vertical reinforcement can be obtained as:

$$T_{u,out} = \frac{\left(f_{yv}A_{sv,out} + \frac{N_s}{2}\right) \cdot 8I_p}{t(2D-t)\pi D_{out}} \cdot \tan\alpha \quad (14)$$

For the horizontal reinforcement, the vertical load induces an initial tensile stress. The total tensile force $F_{v,all}$ in the horizontal reinforcement can be calculated as:

$$F_{h,all} = \frac{Tt(2D-t)}{8I_p} \cdot s_h \cdot \tan\alpha + \frac{\mu_s N_s}{A_{sv}} \cdot A_{sh,1} \quad (15)$$

Where, μ_s is the poisson's ratio of steel reinforcement, taken as 0.3; $A_{sh,1}$ is the sectional area of one horizontal reinforcement bar.

Thus, the torsional capacity contributed by the yielding of the outer horizontal reinforcement is given by:

$$T_{u,out} = \frac{\left(f_{yh}A_{sh,1} - \frac{\mu_s N_s}{A_{sv}} \cdot A_{sh,1}\right) \cdot 8I_p}{t(2D-t)s_h} \cdot \cot\alpha \quad (16)$$

By combining Eqs. (14) and (16), and eliminating $T_{u,out}$, the following equation is obtained:

$$\tan\alpha = \sqrt{\frac{\left(f_{yh}A_{sh,1} - \frac{\mu_s N_s}{A_{sv}} \cdot A_{sh,1}\right) \cdot \pi D_{out}}{\left(f_{yv}A_{sv,out} + \frac{N_s}{2}\right) \cdot s_h}} \quad (17)$$

By substituting Eq. (17) into (16), the torsional capacity $T_{u,out}$ when both the vertical and horizontal reinforcement of out layer yield can be obtained as follows:

$$T_{u,out} = \frac{8I_p}{t(2D-t)} \sqrt{\frac{\left(f_{yh}A_{sh,1} - \frac{\mu_s N_s}{A_{sv}} \cdot A_{sh,1}\right) \cdot \left(f_{yv}A_{sv,out} + \frac{N_s}{2}\right)}{\pi D_{out} \cdot s_h}} \quad (18)$$

The derivation for the torsional capacity when the inner reinforcement yields ($T_{u,in}$) is similar to the above process, and thus will not be repeated. $T_{u,in}$ is calculated by:

$$T_{u,in} = \frac{8I_p}{t(2D-3t)} \sqrt{\frac{\left(f_{yh}A_{sh,1} - \frac{\mu_s N_s}{A_{sv}} \cdot A_{sh,1}\right) \cdot \left(f_{yv}A_{sv,in} + \frac{N_s}{2}\right)}{\pi D_{in} \cdot s_h}} \quad (19)$$

Comparing Eqs. (18) and (19), it can be observed that when the sectional area and strength of the inner and outer layers of reinforcement are the same, $T_{u,out}$ is always less than $T_{u,in}$. This is because the torsional shear flow generated follows a trapezoidal distribution along the thickness of the section (as shown in Fig. 35), and the shear stress in the outer layer is always greater. This indicates that when the configurations of the inner and outer reinforcements are identical, the outer reinforcement will inevitably yield first. The ultimate capacity obtained by considering the yielding of the outer layer reinforcement as the ultimate state will always be smaller. Only when the thickness is significant small, satisfying the Bredt's shear stress distribution, will the ultimate capacities of both layers be close. Considering that in pre-stressed concrete towers, the reinforcement configurations of the inner and outer layers are typically the same, it is conservative to take the torque at the yielding of the outer reinforcement as the torsional capacity of the reinforcement (T_{us}).

The torsional capacity of the reinforcement T_{us} calculated by the above method is compared with the ultimate torque carried by the reinforcement extracted from the FE model ($T_{us,s}$), as shown in Fig. 37. The results show that the current calculated results are generally larger than the simulated results (data with large deviations in the figure correspond to cases of joint failure). This indicates that the theoretical derivation based on the traditional variable-angle truss model is not conservative, and should be revised accordingly. In the above derivation, it was assumed that both the horizontal and vertical reinforcements

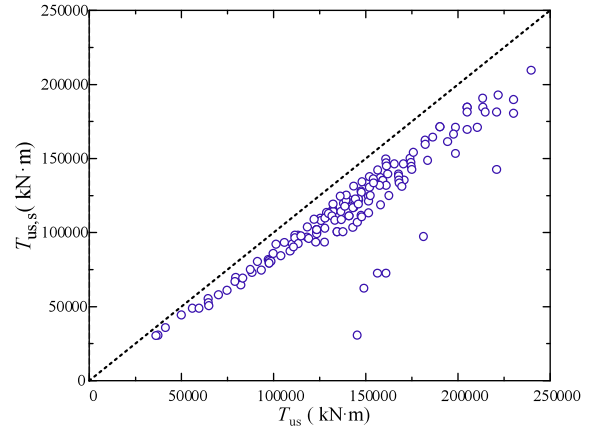


Fig. 37. Comparison between $T_{us,s}$ and T_{us} before modification.

yield at the ultimate state. However, based on the mechanism analysis results, the vertical reinforcement does not generally yield. Assuming full yielding of the vertical reinforcement leads to an overestimation of the theoretical results, making them not conservative. To improve the accuracy, a reduction factor α_{vt} for the yielding of the vertical reinforcement is introduced to modify the theoretical results:

$$T_{us} = \frac{8I_p}{t(2D-t)} \sqrt{\frac{\left(f_{yh}A_{sh,1} - \frac{\mu_s N_s}{A_{sv}} \cdot A_{sh,1}\right) \cdot \left(\alpha_{vt} f_{yv} A_{sv,out} + \frac{N_s}{2}\right)}{\pi D_{out} \cdot s_h}} \quad (20)$$

The following equations can be obtained by transforming Eq. (20) into a linear regression relation:

$$y = \alpha_{vt} x \quad (21)$$

$$\text{With, } y = \frac{\left[\frac{T_{us}t(2D-t)}{8I_p}\right]^2 \cdot \pi D_{out} \cdot s_h}{f_{yh}A_{sh,1} - \frac{\mu_s N_s}{A_{sv}} \cdot A_{sh,1}} - \frac{N_s}{2}$$

$$x = f_{yv} A_{sv,out}$$

By replacing T_{us} with the ultimate torque borne by the reinforcement in the FE model $T_{us,s}$, the regression analysis based on the simulated results can be obtained, as shown in Fig. 38. It can be observed that the ratio of y to x mostly locates within the range of 0.6 to 0.8. It should be noted that for towers with larger axial compression ratios ($m > 0.4$), the failure mode tends to be dominated by concrete compressive failure, resulting in lower utilization of the yield strength of the vertical reinforcement. Excluding the aforementioned samples, the lower bound of the interval can be conservatively taken as the calculation factor, with

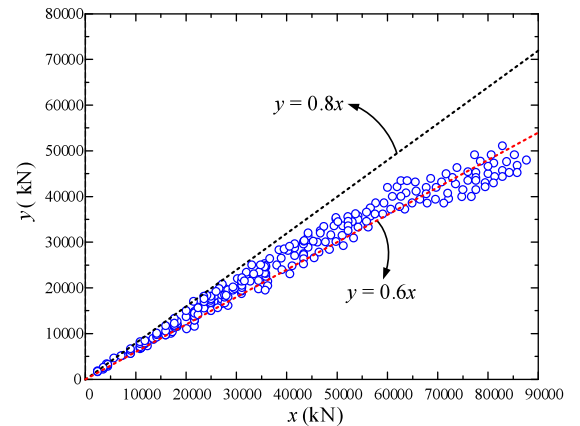


Fig. 38. Regression analysis based on Eq. (21).

the slope of the fitted curve set to 0.6 ($\alpha_{vt} = 6$). Therefore, Eq. (21) can be converted to:

$$T_{us} = \frac{8I_p}{t(2D-t)} \sqrt{\frac{\left(f_{yh}A_{sh,1} - \frac{\mu_s N_s}{A_{sv}} A_{sh,1}\right) \cdot \left(0.6f_{yv}A_{sv,out} + \frac{N_s}{2}\right)}{\pi D_{out} \cdot S_h}} \quad (22)$$

As can be observed from Fig. 37, setting α_{vt} to 0.6 leads to most simulated results $T_{us,s}$ being greater than the calculated torsional capacity of the reinforcement T_{us} , with relatively small errors. Some data points significantly fall below the fitting line $y = 0.6x$ because they correspond to the compressive failure of tower segment, which has been excluded by setting the design limit of axial compression ratio $m \leq 0.4$. Thus, Eq. (22) can be adopted to conservatively estimate the torsional capacity of the reinforcement.

5.1.2. Torsional capacity of the concrete T_{uc}

According to the mechanism analysis results, the torque is primarily borne by the concrete before the principal tensile stress reaches the tensile strength and cracks appear. The contribution of the concrete to the ultimate torsional capacity of the tower is mainly reflected in the cracking torque T_{cr} , and increasing the reinforcement ratio does not significantly increase the cracking torque. Thus, the contribution of the reinforcement is neglected, and only the effect of the concrete is considered. When the maximum shear stress is reached at the out edge of the section, the concrete section may undergo a slight plastic stress redistribution. However, the development of plasticity is not sufficient to allow the entire section to reach the material strength. The actual cracking torque T_{cr} lies between the elastic cracking torque and the plastic cracking torque. In GB50010–2010 [56], the plastic cracking torque is approximately adopted for calculation. To account for the material deviation of concrete, a tensile strength reduction coefficient of

combined with the cracking torque without considering the initial compression to form the concrete's torsional capacity. This can be considered as the contribution of the initial compression to the improvement on the cracking torque. Based on this method, a contribution coefficient α_{tc} of initial compression is introduced, and the term of initial compression contribution is added to the cracking torque given by Eq. (23), obtaining the torsional capacity of concrete T_{uc} :

$$T_{uc} = \frac{2f_t I_p}{D} + \alpha_{tc} \frac{N_t}{A_c} \frac{2I_p}{D} \quad (24)$$

Where, N_t is the total axial force. Based on the results of Section 4.5, a loss factor of 5 % is considered to calculate the total prestressing force at the ultimate state. Thus, N_t can be calculated as $N_t = 0.95P + N$.

By replacing T_{uc} with the ultimate torques borne by the concrete $T_{uc,s}$ in the FE models, the comparison based on simulated results can be obtained, as shown in Fig. 39. Excluding the samples with concrete compressive failure, the lower bound can be conservatively used as the fitting curve, resulting in a contribution coefficient of initial compression $\alpha_{tc} = 0.114$. From this, the torsional capacity of concrete T_{uc} can be obtained:

$$T_{uc} = \frac{2f_t I_p}{D} + 0.114 \cdot \frac{0.95P + N}{A_c} \frac{2I_p}{D} \quad (25)$$

From Fig. 39, it can be seen that when α_{tc} is set to 0.114, the simulated results are quite close to T_{uc} calculated by equation (25). Furthermore, most of the simulated results are greater than the calculated results, indicating that this method provides a relatively conservative prediction of the torsional capacity of concrete. Combining Eqs. (6), (22) and (25), the torsional capacity of the tower segment $T_{u,t}$ governed by the yield of the outer layer reinforcement can be given by:

$$T_{u,t} = T_{uc} + T_{us} = \frac{2f_t I_p}{D} + 0.114 \cdot \frac{0.95P + N}{A_c} \frac{2I_p}{D} + \frac{8I_p}{t(2D-t)} \sqrt{\frac{\left(f_{yh}A_{sh,1} - \frac{\mu_s N_s}{A_{sv}} A_{sh,1}\right) \cdot \left(0.6f_{yv}A_{sv,out} + \frac{N_s}{2}\right)}{\pi D_{out} \cdot S_h}} \quad (26)$$

0.7 is introduced to reduce the plastic cracking torque. However, according to the mechanism analysis, most cross-sections of prestressed concrete towers are thin-walled circular sections. Due to the large hollow ratio, the stress redistribution is weak after the maximum shear stress occurs. After cracking, many positions within the section cannot undergo plastic development. Therefore, for simplicity and conservatism, the elastic cracking torque is taken as the concrete contribution to the ultimate capacity. According to the torsional mechanism, the state when the maximum principal tensile stress σ_{tp} reaching the concrete tensile strength can be considered as the maximum shear stress occurs in the section. If no prestressing force or vertical load is applied to the tower, T_{cr} can be calculated as:

$$T_{cr} = \frac{2f_t I_p}{D} \quad (23)$$

Nevertheless, the prestressed concrete tower is inevitably in a compressive state before experiencing the torque. The initial compression causes the concrete elements to simultaneously bear three-dimensional normal stresses, which results in the principal tensile stress of the concrete no longer being equal to the torsional shear stress. In GB50010–2010, an empirical formula based on experimental results is used to consider the beneficial contribution of the initial compression to the torsional capacity of concrete. This term is the product of the compressive stress and the section's plastic resistance moment, which is

5.2. Ultimate torque of horizontal joint

When the torsional capacity of the tower segment is too large, the tower may exhibit joint failure under torsion. In this case, the ultimate capacity of the tower is governed by the ultimate torque of joint failure $T_{u,j}$. Assuming that the torsional capacity of the joint is entirely provided by the frictional force generated by the vertical load and prestressing force, the mechanism analysis results show that when the torque exceeds the torsional capacity provided by the frictional force, the tower will slip and rotate along the joint, making it unable to bear the torque. The shear stress distribution on the joint interface before joint failure is shown in Fig. 40. The joint interface is constrained by friction, resulting in a frictional shear stress τ_f opposite to the direction of torsion. The resultant of the frictional shear stress at the joint interface is the total frictional force F_f , which balances with the torsional shear force V_t at the bottom of the tower segment. V_t is calculated by:

$$V_t = \int_{-\pi}^{\pi} \int_{\frac{D}{2}}^{\frac{D}{2}} \frac{T}{I_p} \rho \cdot \rho d\rho d\theta = \frac{\pi T t}{6I_p} (3D^2 - 6Dt + 4t^2) \quad (27)$$

The total friction force F_f is equal to the static friction force before the joint slips. After the torsional shear force V_t exceeds the maximum static

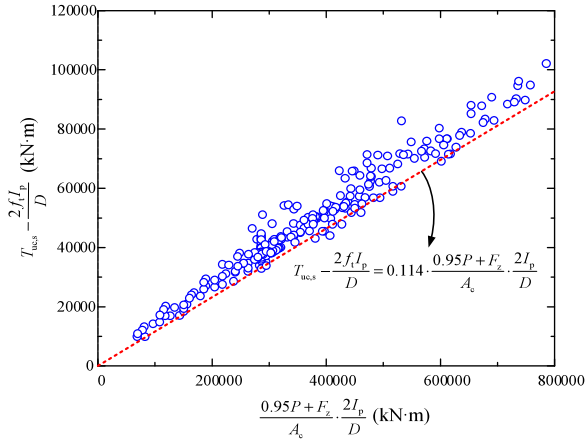


Fig. 39. Regression analysis based on Eq. (24).

friction force $F_{f,max}$, the joint begins to slip and rotate around torsional direction. The maximum static friction force is only related to the total compression on the joint interface. Considering the state when joint begins to slip as the ultimate equilibrium state and ignoring the difference between the maximum static friction force and the sliding friction force, the maximum torsional shear force at joint failure $V_{t,max}$ can be expressed as:

$$V_{t,max} = F_{f,max} = \mu(P + N) \quad (28)$$

At the ultimate equilibrium state, the maximum torsional shear force $V_{t,max}$ still satisfies Eq. (27). Thus, by combining Eqs. (27) and (28), the ultimate torque $T_{u,j}$ of the joint failure can be obtained as:

$$T_{u,j} = \frac{6I_p \cdot \mu(P + N)}{\pi t(3D^2 - 6Dt + 4t^2)} \quad (29)$$

Where, P is the total prestressing force. According to results of Section 4.5, no significant prestress variation can be found for towers with joint failure, thus the prestress reduction can be ignored.

5.3. Ultimate torsional capacity calculation

In summary, the practical calculation method for the ultimate

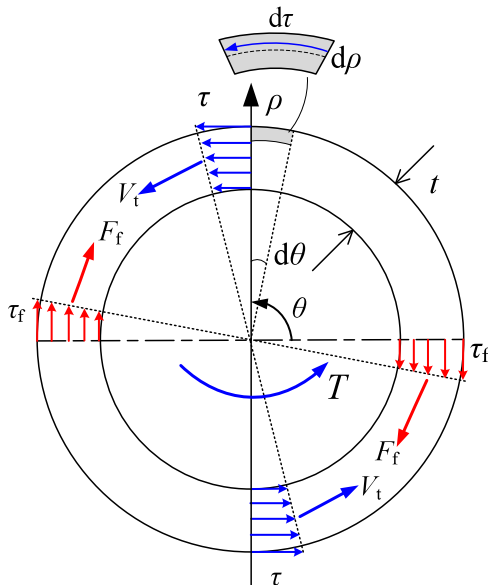


Fig. 40. Shear stress of joint interface.

torsional capacity T_u can be concluded as follows. T_u is controlled by the torsional capacity of tower segment $T_{u,t}$ when the torsional failure of tower segment occurs, and T_u is controlled by the ultimate torque of joint failure $T_{u,j}$. The smaller value of the two should be taken as the calculated ultimate torsional capacity $T_{u,c}$, which can be summarized as:

$$T_{u,c} = \min(T_{u,j}, T_{u,t}) \quad (30)$$

$$\text{With, } T_{u,t} = \frac{2f_t I_p}{D} + 0.114 \frac{0.95P + F_z}{A_c} \frac{2I_p}{D} + \frac{8I_p}{t(2D - t)} \sqrt{\frac{\left(f_{yh} A_{sh,1} - \frac{\mu_s N_s}{A_{sv}} A_{sh,1}\right) \cdot \left(0.6f_{yv} A_{sv,out} + \frac{N_s}{2}\right)}{\pi D_{out} \cdot S_h}}$$

$$T_{u,j} = \frac{6I_p \cdot \mu(P + N)}{\pi t(3D^2 - 6Dt + 4t^2)}$$

This method is only applicable to prestressed concrete towers that do not experience concrete compressive failure. Therefore, based on the results of parametric analysis and mechanism analysis, it is recommended to limit the total axial compression ratio $m \leq 0.4$. To verify the accuracy and applicability of the proposed calculation method, the ultimate torsional capacity T_u obtained from simulated and experimental results were compared with the calculated results $T_{u,c}$, which is illustrated in Fig. 41. It can be seen that the calculated ultimate capacity aligns well with the experimental and simulated results. The error for the experimental result is 2.99 %, while the average calculation error for both the experimental and simulated results is -3.01 %, with a standard deviation (SD) of 0.028. This demonstrate that the calculated results have high accuracy and is generally conservative. In general, the proposed calculation method can effectively and conservatively predict the ultimate torsional capacity of prestressed concrete towers within commonly used parameter ranges, providing guidance for engineering design.

6. Conclusions

This paper presents a comprehensive study on the torsional behavior of wind turbine prestressed concrete tower. Experimental study was conducted and the parameters were extended and discussed through FE models, thereby the full-process torsional mechanism of the tower was revealed. Based on the mechanism, the method for calculating the ultimate torsional capacity were proposed. The following conclusions can be drawn:

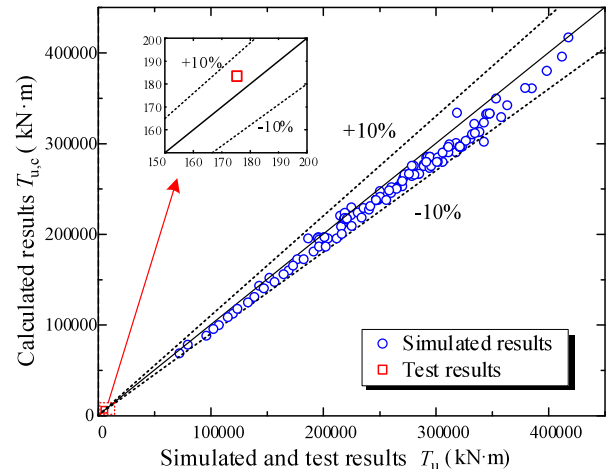


Fig. 41. Comparisons between the calculated results and simulated/test results.

- (1) Under torsion, dense diagonal cracks appeared in the prestressed concrete tower specimen, and severe torsional deformation was observed in the reinforcement between the cracks. The hysteresis curve of the specimen was full, with a significant plastic stage. During the torsion process, the reinforcement exhibited noticeable plastic deformation, resulting in considerable energy dissipation and residual displacement. However, the torsional yielding of the reinforcement could contribute to a good ductility.
- (2) FE models of the prestressed concrete tower were established and validated by test results. The effects of tower dimensions, reinforcement layout, and prestressing layout on the $T-\theta$ curve of the prestressed concrete tower were analyzed. Besides, two failure modes and their corresponding typical $T-\theta$ curves were summarized. When the torsional capacity of the tower segment is low, the torsional failure of the segment will occur. When the torsional capacity of the tower segment is high or the prestress is low, the joint failure will occur.
- (3) The shear stress in the cross-section does not conform well to Bredt's shear stress distribution but generally aligns with Saint-Venant's torsion theory. After cracking, the concrete develops diagonal cracks, dividing it into multiple oblique compression strips, where the concrete between cracks can still bear tensile stress at the ultimate state. Increasing the prestress ratio can delay the cracking, but an excessively high prestress ratio may lead to brittle compressive failure.
- (4) For a tower with reasonable reinforcement and prestress, the ultimate state is governed by the yielding of the reinforcement. At the ultimate state, the reinforcement and the cracking concrete form a cooperative torsional system, both being able to participate in tension. The horizontal reinforcement can be yielding, while the strength of vertical reinforcement cannot be fully utilized. Increasing the prestress ratio significantly enhances the torsional capacity provided by the concrete, while reducing the contribution of the reinforcement.
- (5) The shear force of the joint increases with the torque but does not exceed the maximum static friction force of the interface. Higher prestress increases the maximum static friction force of the interface, making the joint less prone to failure. The prestress decreases continuously as the torsional angle increases, and at the ultimate state, the prestress reduction is significantly affected by the failure mode and the prestress ratio. A loss factor of 5 % is adopted to calculate the total prestress at the ultimate state. But when joint failure occurs, the reduction of prestress can be ignored.
- (6) Based on the torsional mechanism of prestressed concrete towers, calculation methods for the torsional capacity of the tower segment and the ultimate torque of joint failure were established using Saint-Venant's torsion theory and the variable-angle truss model. These methods can be used to calculate the ultimate torsional capacity in conjunction with the torsional failure mode of tower.

Considering the critical interaction between bending behavior and torsional behavior, it is recommended to subsequently conduct more experimental and numerical studies on the combined bending-torsion performance of the tower based on the results of this paper. From this, the comprehensive design method for the prestressed concrete towers can be obtained.

CRediT authorship contribution statement

Wei Ren: Writing – review & editing, Visualization, Investigation, Formal analysis. **Ran Deng:** Visualization, Validation, Funding acquisition. **Xu-Hong Zhou:** Resources, Project administration, Conceptualization. **Yu-Hang Wang:** Supervision, Resources, Funding acquisition. **Yu-Sen Liu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by National Key Research and Development Program (2024YFF0505400), National Natural Science Foundation of China (52308212), Fundamental Research Funds for the Central Universities (2024CDJYXTD-005), Chongqing Science and Technology Bureau (2024NSCQ-JQX0096).

Data availability

No data was used for the research described in the article.

References

- [1] W. Ren, R. Deng, X.H. Zhou, et al., Compressive behavior of the steel-concrete composite adapter for wind turbine hybrid towers, *Eng. Struct.* 280 (2023) 115703.
- [2] R. Deng, X.H. Zhou, et al., Experimental study on tapered concrete-filled double skin steel tubular columns under torsion, *Thin-Walled Struct.* 177 (2022) 109444.
- [3] J. Lana, P. Júnior, C.A. Magalhes, et al., Behavior study of prestressed concrete wind-turbine tower in circular cross-section, *Eng. Struct.* 27 (2021) 111403.
- [4] J. Chen, J. Li, X. He, Design optimization of steel-concrete hybrid wind turbine tower based on improved genetic algorithm, *Struct. Des. Tall Spec. Build.* 29 (2020) e1741.
- [5] Z. Peng, C. Shukang, G. Bin, Simulation and optimization design of tower structure of wind generating set, in: *International Conference on Advanced Computer Theory and Engineering* 4, 2010, pp. 309–312.
- [6] H.W. Ma, R. Meng, Optimization design of prestressed concrete wind-turbine tower, *Sci. China-Technol. Sci.* 57 (2) (2014) 414–422.
- [7] Y. Yao, Analysis on Load Capacity and Optimization of Fabricated Steel-Concrete Hybrid Tower, Nanjing University of Aeronautics and Astronautics, Nanjing, 2019.
- [8] F. Klein, F. Füll, T. Betz, et al., Experimental study on the joint bearing behavior of segmented tower structures subjected to normal and bending shear loads, *Struct. Concr.* 23 (2022) 1370–1384.
- [9] C. Von Der Haar, S. Marx, Design aspects of concrete towers for wind turbines, *J. S. Afr. Inst. Civ. Eng.* 57 (4) (2015) 30–37.
- [10] C. Kang, S. Hartwig, S. Marx, Behavior of externally prestressed segmental towers' dry joint under torsion effects, *Struct. Concr.* 20 (2019) 1350–1357.
- [11] J. Chen, W. Lin, J. Li, Ultimate torsional moment of dry horizontal joint for prefabricated concrete tower, *Struct. Des. Tall Spec. Build.* 33 (15) (2024) e2156.
- [12] A.A. Gvozdev, N.N. Lessig, L.K. Rulle, Research on reinforced concrete beams under combined bending and torsion in the Soviet Union, *Spec. Publ.* 18 (1950) 307–336.
- [13] E. Rausch, *Design of Reinforced Concrete in Torsion (Berechnung Des Eisenbetons gegen Verdrehung)*, Springer, Berlin, 1938.
- [14] B. Thurlimann, Torsional strength of reinforced and prestressed concrete beams-CEB approach, *Spec. Publ.* 59 (1979) 117–144.
- [15] M. Collins, Torque-twist characteristics of reinforced concrete beams, *Inelast. Non-Linearity Struct. Concr.* (1973) 211–232.
- [16] T.T. Hsu, Y.L. Mo, Softening of concrete in torsional members-theory and tests, *J. Proc.* 82 (3) (1985), 290–303.
- [17] T.T. Hsu, Unified approach to shear analysis and design, *Cem. Concr. Compos.* 20 (6) (1998) 419–435.
- [18] C.H. Jeng, T.T.C. Hsu, A softened membrane model for torsion in reinforced concrete members, *Eng. Struct.* 31 (9) (2009) 1944–1954.
- [19] C.H. Jeng, Unified softened membrane model for torsion in hollow and solid reinforced concrete members: modeling precracking and postcracking behavior, *J. Struct. Eng.* 141 (10) (2015) 04014243.
- [20] R.S. Stramandinoli, R. La, L. H., An efficient tension-stiffening model for nonlinear analysis of reinforced concrete members, *Eng. Struct.* 30 (7) (2008) 2069–2080.
- [21] L.F.A. Bernardo, J.M.A. Andrade, S.M.R. Lopes, Modified variable angle truss-model for torsion in reinforced concrete beams, *Mater. Struct.* 45 (2012) 1877–1902.
- [22] L.F.A. Bernardo, J.M.A. Andrade, N.C.G. Nunes, Generalized softened variable angle truss-model for reinforced concrete beams under torsion, *Mater. Struct.* 48 (2015) 2169–2193.
- [23] L.F.A. Bernardo, M.M. Teixeira, Modified softened truss-model for prestressed concrete beams under torsion, *J. Build. Eng.* 19 (2018) 49–61.
- [24] H. Becks, M. Schmidt, S. Bosbach, et al., Large-scale combined torsional and axial loading system (TorAx)—A new perspective on multiaxial testing of fracture and stress transfer in plain and reinforced concrete, *Eng. Struct.* 302 (2024) 117391.
- [25] H.R. Valipour, S.J. Foster, Nonlinear reinforced concrete frame element with torsion, *Eng. Struct.* 32 (4) (2010) 988–1002.
- [26] H. Ju, D. Lee, et al., Maximum torsional reinforcement ratio of reinforced concrete beams, *Structures* 23 (2020) 481–493.
- [27] H. Ju, D. Lee, et al., Minimum torsional reinforcement ratio for reinforced concrete members with steel fibers, *Compos. Struct.* 207 (2019) 460–470.

- [28] C.H. Jeng, M. Chao, Unified rational formula for pre-cracking torsional stiffness of solid and hollow reinforced concrete members, *Eng. Struct.* 99 (2015) 92–107.
- [29] A. Kuan, E.P. Bruun, E.C. Bentz, M.P. Collins, Nonlinear sectional analysis of reinforced concrete beams and shells subjected to pure torsion, *Comput. Struct.* 222 (2019) 118–132.
- [30] C.H. Jeng, M. Chao, H.C. Chuang, Torsion experiment and cracking-torque formulae of hollow prestressed concrete beams, *Eng. Struct.* 196 (2019) 109325.
- [31] Y.M. You, A. Belarbi, Thickness of shear flow zone in a circular RC column under pure torsion, *Eng. Struct.* 33 (9) (2011) 2435–2447.
- [32] T.G. Mondal, S.S. Prakash, Effect of tension stiffening on the behaviour of reinforced concrete circular columns under torsion, *Eng. Struct.* 92 (2015) 186–195.
- [33] X. Cao, W.J. Zhang, Y.C. Ren, et al., Torsional capacity of ultra-high-performance concrete beams using rectangle stirrup, *J. Build. Eng.* 69 (2023) 106231.
- [34] J. Zhou, C. Li, Z. Feng, D.Y. Yoo, Experimental investigation on torsional behaviors of ultra-high-performance fiber-reinforced concrete hollow beams, *Cem. Concr. Compos.* 129 (2022) 104504.
- [35] J. Zhou, C. Li, D.Y. Yoo, J. He, et al., Modified softened membrane model for ultra-high-performance fiber-reinforced concrete solid and hollow beams under pure torsion, *Eng. Struct.* 270 (2022) 114865.
- [36] C. Li, J. Zhou, L. Ke, S. Yu, et al., Failure mechanisms and loading capacity prediction for rectangular UHPC beams under pure torsion, *Eng. Struct.* 264 (2022) 114426.
- [37] M. Abdoli, D. Mostofinejad, M. Eftekhar, Ultimate torsional resistance and failure modes of FRP-strengthened reinforced concrete members: a nonlinear design model, *Eng. Struct.* 283 (2023) 115867.
- [38] M. Said, A. Salah, A. Erfan, A. Esam, Experimental analysis of torsional behavior of hybrid fiber reinforced concrete beams, *J. Build. Eng.* 71 (2023) 106574.
- [39] A. Karimipour, et al., Torsional behaviour of rectangular high-performance fibre-reinforced concrete beams, *Structures* 35 (2022) 511–519.
- [40] R.F. Hassan, et al., Experimental study and theoretical prediction on torsional strength with different steel fiber reinforced concretes and cross-section areas, *Eng. Struct.* 251 (2022) 113559.
- [41] J.T. Hewes, M.J.N. Priestley, Seismic Design and Performance of Precast Concrete Segmental Bridge Columns, University of California at San Diego, San Diego, CA, 2002. Report No. SSRP-2001/25.
- [42] Z.Y. Bu, Y.C. Ou, J.W. Song, et al., Cyclic loading test of unbonded and bonded posttensioned precast segmental bridge columns with circular section, *J. Bridge Eng.* 21 (2) (2016) 04015043.
- [43] W. Ren, X.H. Zhou, R. Deng, et al., Compression-bending behavior of thin-walled prestressed concrete tower with horizontal joint for wind turbines: experimental study and calculation model, *Thin-Walled Struct.* 211 (2025) 113154.
- [44] GB/T 50081-2019, Standard for test methods of concrete physical and mechanical properties, Ministry of Housing and Urban-Rural Development of PRC, Beijing (China), 2019.
- [45] GB/T 228.1–2010, Metallic materials–Tensile testing–Part 1: method of test at room temperature, Standardization Administration of China, Beijing(China), 2010.
- [46] J.G. Nie, Y.H. Wang, J.S. Fan, Experimental study on seismic behavior of concrete filled steel tube columns under pure torsion and compression–torsion cyclic load, *J. Constr. Steel Res.* 79 (2012) 115–126.
- [47] Y.H. Wang, W. Wang, J. Chen, Seismic behavior of steel tube confined RC columns under compression-bending-torsion combined load, *J. Constr. Steel Res.* 143 (2018) 83–96.
- [48] GJGT 101-2015, Specification for Seismic Test of Buildings, Ministry of Housing and Urban-Rural Development of PRC, Beijing (China), 2015.
- [49] J.T. Hewes, M.J.N. Priestley, Seismic Design and Performance of Precast Concrete Segmental Bridge Columns, University of California at San Diego, San Diego, CA, 2002. Report No. SSRP-2001/25.
- [50] Y.C. Ou, Precast Segmental Post-Tensioned Concrete Bridge Columns for Seismic Regions, State University of New York at Buffalo, Buffalo, 2007.
- [51] Z.Y. Bu, Y.C. Ou, Simplified analytical pushover method for precast segmental concrete bridge columns, *Adv. Struct. Eng.* 16 (5) (2013) 805–822.
- [52] Z.Q. Wang, Z.W. Yu, Concrete damage model based on energy loss, *J. Build. Mater.* (04) (2004) 365–369.
- [53] E. Hognestad, N.W. Hanson, D. McHenry, Concrete stress distribution in ultimate strength design, *ACI Struct. J.* 7 (1960) 1–28.
- [54] L.H. Han, G.H. Yao, Z. Tao, Performance of concrete-filled thin-walled steel tubes under pure torsion, *Thin-Walled Struct.* 45 (1) (2007) 24–36.
- [55] CEB-FIP Model Code 90, Thomas Telford Ltd, London, 1993.
- [56] GB/50010-2010, Code for Design of Concrete Structures, China Construction Industry Press, Beijing (China), 2015.
- [57] Wind Energy Generation Systems - Part 6: Tower and Foundation Design requirements, IEC 61400-6-2020, International Electrotechnical Commission, Geneva, 2020.